

Tense and perspective: A solution to the ‘*then*’-present puzzle*

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Abstract

In this paper, we focus on the cross-linguistic incompatibility between the temporal adverbial ‘*then*’ and the present tense. We show that the pattern holds in both matrix and embedded environments, across an array of languages (including Modern Greek, Modern Hebrew, Russian and Japanese) each behaving differently with respect to tense shift. To account for this incompatibility, we propose that the interpretation of tenses is sensitive to a perspective parameter. Tense shift occurs thanks to a perspective-shifting operator, similar to operator-based analyses of indexical shift (Anand and Nevins 2004; Anand 2006; Deal 2020). The main advantage of this account over alternative theories of tense shift, such as relative tense (Ogihara, 1996, 2022) or tense deletion (Ogihara and Sharvit, 2012), is that we are able to distinguish between shifted and deleted tense. This is needed to explain the incompatibility of ‘*then*’ with the shifted present but not with deleted past. Finally, we argue that the newly introduced perspective parameter needs to be separate from the speech context and the evaluation index.

Keywords: Tense shift, Present, Deleted tense, Perspectives, Then, Temporal adverbials

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1 Introduction

Tense is often thought of as a deictic category, establishing a *temporal reference* relative to a **temporal anchor**—canonically the time of utterance (Comrie 1985). In other words, it often tells us when something happened with respect to the time of utterance. For example, the present usually establishes a temporal reference that *overlaps* with its temporal anchor, while the past tense establishes a reference that *precedes* it. While much work focuses on the nature of the temporal reference, relatively little attention has been paid to the temporal anchor, and its formal status lacks adequate independent empirical motivation. For instance, in an *intensional* system (following Prior 1957, 1967) where tenses manipulate meta-linguistic parameters, i.e., time *indices*, the temporal anchor is the initial time index. On the contrary, in an *extensional* system (following Partee 1973) where temporal reference is modeled as a time *variable* in the object language, the temporal anchor corresponds to a (covert) variable.¹

This paper takes an initial step towards clarifying the formal status of temporal anchors. We argue that although the *reference* of tense can be flexible—and is thus characterized as a time variable—the temporal anchor behaves in a more constrained manner. Within a minimal clausal domain, the anchor has to be fixed, behaving like an evaluation index and unlike a variable. At the same time, we will show that the temporal anchor cannot simply be identified with other familiar parameters that specify the circumstances of the utterance, such as the evaluation index and the context parameter. The proposal is thus to model the temporal anchor as a distinct interpretation parameter, which we call the *temporal perspective*. Our main empirical motivation comes from the interaction between tense and the temporal adverbial ‘*then*’ across a number of languages, and across various embedding structures.² In the rest of the introduction, we will briefly expand on our empirical focus, identify the empirical problem, and provide an overview of the proposal.

1.1 The ‘*then*’-present puzzle

The English expression *then* has a variety of usages. This paper focuses on the use of *then* (and its counterparts in other languages) as a standard temporal adverbial as exemplified in (1). Descriptively, the adverbial *then* specifies the time of the event/state depicted by the adjoining VP.

- (1) The janitor turned off the lights. The room was empty *then*. (Anand and Toosarvandani 2019: (40))

Just like tenses, the temporal adverbial ‘*then*’ is commonly assumed to be a deictic expression, forcing a *distal* reference relative to its temporal anchor (Schiffrin 1992). That is, if it is anchored to the time of utterance (as in (1)), *then* can only refer to a time in the past or future. In addition, the reference of the

¹In general, extensional systems containing object-language time variables have been argued to have the necessary expressive power to model natural language. Many of the arguments involve temporal expressions such as English *then* (and *now*, see e.g. Cresswell 1990). In this spirit, we adopt an extensional system, modeling the temporal reference of tenses with object language time variables. But overall, the choice is not crucial for our main arguments (see also Cresswell 1990 for an equally expressive intensional system).

²Throughout this paper, we will use the corner quoted ‘*then*’ as a placeholder for temporal adverbials across languages corresponding to English *then*. It will refer to adverbials as opposed to periphrastic anaphoric expressions (e.g. in Modern Greek, ‘*then*’ corresponds to *tote* ‘then’, as in (11), as opposed to *ekino ton kero* ‘at that time’).

adverbial *then* is often anaphoric. In (1), for example, *then* refers to a time established in the previous sentence, i.e. the time the janitor turned off the lights.

However, even when focusing on the temporal uses of *then*, we find that it has various semantic and discourse-level contributions (Schiffrin 1992; Anand and Toosarvandani 2019). Indeed, compare clause-final *then* with clause-initial *then*; the contrast in (2) shows that while the former simply refers to an antecedent time, the latter may refer to a time *after* the antecedent, in coherence with the narrative progression.³

- (2) a. #People began to leave. The room was empty *then*.
 b. People began to leave. *Then*, the room was empty. (Anand and Toosarvandani 2019: (36))

That said, in this paper, we simply use the deictic component of the adverbial *then* as a probe into the semantics of tense and of relevant syntactic dependency mechanisms. Therefore, instead of multi-sentence discourses as in (1) and (2), we focus on single-sentence utterances with embedding structures. We restrict ourselves to cases where *then* occurs in the clause-final (or clause-internal) position of a tensed embedded clause, as illustrated in (3) (see also Ogihara and Sharvit (2012); Vostrikova (2019); Tsilia (2021); Tsilia and Zhao (2025)). Therefore, in terms of reference, *then* is always anaphoric to a matrix temporal expression (here *two years ago*) or the matrix tense.

- (3) Two years ago, John knew that Mary was pregnant *then*.

For the same reason, we will set aside the conditional/epistemic use of *then* illustrated in (4), as well as other usages of discourse marking as in (5), since they lack a clear, temporally deictic component, and arguably do not contribute to the temporal semantics. Indeed, (4) and (5) below are compatible with a reading of the *then*-clause as referring to the time of utterance, and being simultaneous with its temporal antecedent. We set aside these cases, and focus instead on clearly deictic usages of *then*.

- (4) If Peter doesn't want to run for president, *then* I don't know what we are gonna do.
 (adapted from Iatridou 1993)

³Schiffrin and Anand and Toosarvandani disagree on the source of the narrative time shift. For Schiffrin, the shift is the result of the discourse function of *then*. For Anand and Toosarvandani, it is simply a general tendency of narrative sequencing. To show this, Anand and Toosarvandani point out that, occasionally, *then* can achieve a backward shift in the discourse. For example, (3) can be understood as Cohen's enjoying Trump's support precedes his incurring his wrath.

- (1) How things have changed since the campaign! Cohen plead guilty last week and incurred Trump's wrath. *Then*, he enjoyed his full support. (Anand and Toosarvandani 2019: (37))

Importantly, they showed that this reading is contingent on the presence of a salient temporal antecedent in the past (in (3) it is the time of the campaign). Otherwise, the time of the narrative has to shift forward. This is demonstrated below by reducing (3) to the two-sentence discourse below.

- (2) Cohen plead guilty last week and incurred Trump's wrath. *Then*, he enjoyed his full support.
 Can only mean: Cohen enjoyed Trump's full support *after* he incurred Trump's wrath. (Anand and Toosarvandani 2019: (38))

In any case, in this paper we focus exclusively on clause-final *then*. Thus, this debate is orthogonal to our arguments.

- (5) There is a cop up the next block. There's another cop over there. *Then* there are two more around the corner. (adapted from Schiffrin 1992: (19))

Now to our main puzzle. To begin with, the temporal adverbial *then* cannot co-occur with a present tense, as shown by the contrast between (6) and (7).⁴

- (6) Mary is feeling sick (*then).
 (7) a. Mary was feeling sick (then).
 b. Mary will be feeling sick (then).

The contrast in these basic cases is not so hard to explain. The present in simple root clauses typically denotes a time overlapping with the time of utterance, while the past and the future denote times different from it. Given the deictic condition that the adverbial *then* has to refer to a time disjoint from the time of utterance, the unacceptability of (6) follows; indeed, we cannot use *then* to refer to a time overlapping with Mary's *current* illness. Accordingly, the examples in (7) are acceptable because both the past and the future denote times that can be restricted by the temporal adverbial *then*.

Things become more interesting when we turn to embedded tenses. In languages such as Modern Greek, Modern Hebrew, Japanese and Russian, an embedded present can receive a shifted interpretation that does not overlap with the time of utterance. The Modern Greek sentence in (8) by default means that Maria's pregnancy overlaps in time with Yanis' past knowledge in 2000. In other words, the embedded present is used to refer to a past time that overlaps with the time of the matrix attitude. We will refer to this phenomenon as *tense shift*, and refer to a present tense with a shifted interpretation as a *shifted present*.

- (8) To 2000, o Yanis **iksere** oti i Maria **ine** egkios.
 The 2000, _{DET} Yanis **know.PAST** that the Maria **be.PRES** pregnant
 'In 2000, Yanis knew that Maria was (LIT. is) pregnant.' Modern Greek

The explanation for the incompatibility between 'then' and the present in root clauses is now too simplistic to be carried over to embedded clauses. If 'then' is only required to denote a time disjoint from the time of *utterance*, then a shifted present should in principle be compatible with it. However, as first noticed by Ogihara and Sharvit (2012) for Modern Hebrew, Vostrikova (2019) for Russian and Sharvit (2018) and Tsilia (2021) for Modern Greek, this is not the case:

- (9) Lifney alpayim šana, Yosef **xašav** še Miriam **ohevet** oto (***az**).
 Before 2,000 year, Yosef **think.PAST** that Miriam **love.PRES** him **then**
 '2,000 years ago, Yosef thought that Miriam loved (LIT. loves) him then.'
 Modern Hebrew (Ogihara and Sharvit 2012: fn. 6)

⁴This paper focuses on the *temporal* adverbial *then* and the ordinary present. We do not discuss the historical present, for which we refer the reader to Anand and Toosarvandani (2020, 2022).

- (10) V 2016 godu Tanja skazala, čto Putin (***togda**) ∅ prezident Rossii.
 in 2016 year Tanja say.PAST that Putin **then** **be.PRES** president.NOM Russia.GEN
 Intended: ‘In 2016 Tanja said that Putin was (LIT. is) the president of Russia then.’

Russian (Vostrikova 2019: (4))

- (11) To 2000, o Yanis **iksere** oti i Maria **ine** egkios (***tote**).
 The 2000, DET Yanis **know.PAST** that the Maria **be.PRES** pregnant **then**
 Intended: ‘In 2000, Yanis knew that Maria was (LIT. is) pregnant then.’

Modern Greek (Tsilia 2021: (78))

Despite the fact that in these languages the present is shiftable and does not necessarily denote the time of utterance, *then* is still incompatible with it. This incompatibility constitutes the central puzzle addressed in this paper, and we will call it the ‘*then*’-present puzzle (see (12)). The overarching goal of this paper is to use the ‘*then*’-present puzzle as a window for a better understanding of tense semantics and tense shift.

(12) The ‘*then*’-present puzzle

The temporal adverbial ‘*then*’ cannot be adjoined to a present-tensed verb phrase, regardless of whether the present tense is shifted or not.

In previous literature, the puzzle has mostly been discussed for present-under-past sentences (Ogihara and Sharvit, 2012; Vostrikova, 2019). In this paper, we further consider present-under-future sentences, and show that the puzzle largely persists (Section 2.2). More importantly, we also compare the behavior of ‘*then*’ across present-under-past and past-under-past sentences, which have important theoretical consequences (Section 3).

Before exploring the puzzle in more detail, it is worth pointing out that prior literature has already refuted the null hypothesis that the incompatibility between ‘*then*’ and the present is due to morphosyntactic reasons. Tsilia, 2021, Section 4.1.2 showed that ‘*then*’ is compatible with present tense morphology in historical present contexts in Modern Hebrew, French, and Modern Greek. It is also compatible with a quantified present in Russian.⁵ As we will also see later on in the paper, English *then* is compatible with the present under future. What is more, English *then* is also compatible with the present when the latter is used to express a future plan (see fn.25). Finally, in Appendix A we present data from present-tensed modals, showing that they are compatible with ‘*then*’ despite their morphology. In this paper, therefore, we will look to solve the puzzle through the semantics of tense and ‘*then*’.

1.2 Overview of the proposal

As indicated above, our investigation into the ‘*then*’-present puzzle is based on an established typology of *tense shift* (Ogihara and Sharvit 2012; Tsilia 2021). Previous literature offers two main structural analyses

⁵We refer the interested reader to Section 4.1.2. in Tsilia (2021), ex.(89)-(98), p.44-46.

for tense shift.⁶ The first one, exemplified by Ogihara (1996, 2022), argues that tense shift results from the shift the temporal anchor. In that view, a shifted present is anchored to a time distinct from the time of utterance, and for (8), it is the time of the matrix attitude. To implement this idea, the temporal anchor is treated as a (covert) pronoun, analogous to the temporal reference. Every tense morpheme therefore carries two indices, one for its reference and one for its anchor. Accordingly, (8) corresponds to the following LF where the anchor of the shifted present is bound by the reference of the matrix past.

- (13) Yanis knew that Maria is pregnant. (Lit. from Modern Greek)
 Yanis know-**PAST**_{1,0}^{ANCHOR} [. . . Maria be-**PRES**_{2,1}^{ANCHOR} pregnant] (to be rejected)

The second approach, exemplified by e.g. von Stechow (2003) and Ogihara and Sharvit (2012), proposes that a shifted present is semantically equivalent to a *deleted* tense (or a *zero* tense, *per* Kratzer 1998; von Stechow 2003). The term ‘deleted tense’ usually refers to an instance of a past tense that gives rise to a simultaneous reading when embedded under a matrix past, as if the past tense feature is no longer interpreted; it is thus in a sense ‘deleted’. Among other languages, the deleted past has been observed in English and Modern Greek (see (14) and (15)). The exact implementation of the ‘deletion’ mechanism differs among theories (we come back to this shortly, see Section 3, fn. 11). Irrespective of the deletion mechanism we use, given that deleted tense gives rise to the same simultaneous reading as the shifted present, it is tempting to analyze them both as involving an uninterpreted tense feature. This idea is implemented in the LF in (15).

- (14) John found out that Mary **was** drunk.
 John find-out-**PAST**_{1,0} [. . . Maria be-**PAST**₁ drunk]
 ~→ Mary was drunk at the time when John found out.
- (15) Yanis knew that Maria is pregnant. (Lit. from Modern Greek)
 Yanis know-**PAST**_{1,0} [. . . Maria be-**PRES**₁ pregnant] (to be rejected)

In this paper, we will reject both of these options, and adopt the LF in (16). According to (16), the temporal anchor is no longer modeled as a pronoun, but as an interpretation **parameter** π which we call the temporal **perspective**. In other words, tenses only bear a single index corresponding to their temporal reference, and tense features restrict the range of this temporal reference with respect to π . Meanwhile, tense shift is achieved through a perspective-shifting operator **OP** _{π} , which takes scope over the entire embedded clause, and works by changing the value of π used to interpret the embedded clause.

⁶A pragmatic account in terms of *cessation implicatures* has been proposed in Altshuler and Schwarzschild (2013b). This account derives the back-shifted reading of past-under-past sentences as a cessation implicature, whenever a viable present tense alternative is available. However, as discussed in a footnote in Altshuler, 2016, p.136 and in Tsilia, 2021, fn.18, p.23, Modern Greek has been shown to be problematic for this account. This is because past-under-past sentences in Modern Greek have a simultaneous reading, despite a present tense alternative being available, suggesting that a cessation implicature is not triggered in this case. Given that pragmatic accounts are hard to parametrise across languages (since we cannot argue that Modern Greek does not draw the implicature) and that in the present paper we are explicitly concerned with Modern Greek among other languages, we do not delve deeper into this account.

(16) Yanis knew that Maria is pregnant.

(Lit. from Modern Greek)

$\llbracket \text{Yanis know-}\mathbf{PAST}_1 \text{ } \llbracket \mathbf{OP}_\pi \text{ Maria be-}\mathbf{PRES}_2 \text{ pregnant} \rrbracket \rrbracket \dots \overset{\text{ANCHOR}}{\pi} \dots$

The preference for (16) is directly motivated by the interactions between tenses and $\llbracket \textit{then} \rrbracket$. (16) is similar to (13) in attributing tense shift to the shift of the anchor. However, treating the anchor as an interpretation parameter as in (16) has the advantage of granting it a ‘global’ effect on all expressions within the domain of interpretation. Together with the fact that $\mathbf{OP}_\pi$ applies only to a propositional node, it guarantees that the tense and $\llbracket \textit{then} \rrbracket$ within the same minimal clause always shift together. This allows (16) to generate only a subset of readings that (13) does, which is a crucial step towards solving the $\llbracket \textit{then} \rrbracket$ -present puzzle. We will expand on this in Section 2.3. Readers familiar with the literature on **indexical shift** may also recognize the parallel between (16) and the theory developed in Anand and Nevins (2004) and Anand (2006). Indeed, as will become clear in Section 4, our theory draws heavily on their analysis of the **SHIFT TOGETHER** effect of indexicals.

As for the second approach as in (15), by identifying the shifted present with deleted tense, it makes the prediction that $\llbracket \textit{then} \rrbracket$ should behave the same with both types of tenses. In Section 3 we will show that this is empirically wrong. In fact, unlike with the shifted present, $\llbracket \textit{then} \rrbracket$ is perfectly compatible with a deleted tense. Consequently, a distinction needs to be made between deleted and shifted tense. To our knowledge, this is the first empirical evidence demonstrating that shifted tense differs from deleted tense. In Section 6, we show that the analysis in (16) can be extended to capture this difference – while the shifted present remains sensitive to a shifted perspective, a deleted tense is genuinely devoid of perspective sensitivity.

Finally, we will argue that the temporal perspective parameter π should be distinguished from other interpretation parameters, such as the utterance context and the evaluation index. Identifying π with the utterance context will predict that tense shift is concomitant with (at least some instances of) indexical shift. We will show that this is not the case. In fact, most of the embedding environments in the languages surveyed in the paper don’t allow indexical shift at all.⁷ Identifying π with the evaluation index is also unattainable because it predicts obligatory shifting of tense and $\llbracket \textit{then} \rrbracket$, which is contrary to fact. Additionally, collapsing π with the evaluation index would no longer let us distinguish between deleted tense and the shifted present, a distinction which we need to explain the compatibility of $\llbracket \textit{then} \rrbracket$ with deleted, but not with shifted tense.

In this paper, we will only delve into the temporal component of the perspective parameter, although there is definitely more to explore in terms of other ‘perspective-sensitive items’, see e.g. Partee (2004), Oshima (2005), McCready (2007), Pearson (2013), Bylina (2014), Sudo (2015), Sundaresan (2018) and many others. In fact, our formalism closely resembles the one in Sudo’s (2015) analysis of locative perspectives and the semantics of motion verbs. Exploring connections between temporal and other kinds of perspective-

⁷Relatedly, a reviewer asks if we could collapse the current system into the bi-contextual systems motivated independently by Free Indirect Discourse (FID) (Banfield, 1982; Doron, 1991; Schlenker, 2004; Sharvit, 2008). But note that FID does not allow for the shifting of tenses and person indexicals. Therefore, bi-contextual systems for FID separate tense and person indexicals on the one hand, and other types of indexicals on the other hand. As a result, they cannot distinguish between the context-sensitivity of tenses and person indexicals. Yet, none of the tense-shifting environments discussed in this paper allow for the shift of person indexicals. Thus, we cannot use a system that treats them on a par.

sensitive expressions is an exciting direction for future research.

The rest of the paper is structured as follows. Section 2 lays out the detailed empirical picture presenting the ‘*then*’-present puzzle. It shows that among a number of languages that constitute a complex typology of tense shift, the puzzle arises consistently with both non-shifted and shifted present. This helps us clarify the point that the temporal anchor is better characterized as an interpretation parameter than a pronoun. Section 3 elaborates on deleted tense, demonstrating that it cannot be equivalent to the shifted present. Section 4 develops a compositional fragment that builds on the theory of indexical shift, which is then used in Section 5 to offer a formal analysis of the ‘*then*’-present puzzle. The account also embodies a new analysis of tense shift across languages. We then show in Section 6 how deleted tense differs from shifted present in our system since the latter but not the former is sensitive to a temporal perspective parameter. Section 7 argues for the necessity of the distinct perspective parameter by showing that it cannot be reduced to the context or the evaluation index. Section 8 concludes.

2 The shifted present and the ‘*then*’-present puzzle

The languages surveyed in this paper include English, Modern Greek, Modern Hebrew, Japanese and Russian. These are all languages with overt tense morphology, and are heavily represented in the typological studies of tense embedding (cf. Ogiwara and Sharvit 2012). In this Section we focus on the (embedded) present tense. Depending on the tense of the main clause, and on the nature of the embedded clause (i.e. whether it is a complement of a speech/attitude verb or a relative clause), languages may or may not shift the embedded present. Subsections 2.1 and 2.2 go through each configuration, starting with the present embedded under *past*-tensed matrix verbs and moving on to the present embedded under *future*. The upshot is that, regardless of whether the present shifts, it is almost always incompatible with the adverbial ‘*then*’. Based on this observation, Subsection 2.3 then discusses why treating temporal anchors as interpretation parameters as in (16) should be the favored strategy over treating them as pronouns, as sketched in (13).

2.1 Present-under-past

When embedded under a past-tensed matrix verb, the present tense in English never shifts. The event(uality) in the embedded clause is always interpreted as occurring at the time of utterance:

- (17) In his childhood, Joseph **met** a woman who **loves** traveling
 ~> the woman loves traveling *at the time of utterance* (Ogiwara and Sharvit 2012: (19a))
- (18) A month ago, John **found** out that Mary **loves** him.
 ~> Mary loves John *at the time of utterance* (Ogiwara and Sharvit 2012: (15a))

In a relative clause, there is no *simultaneous* reading between the matrix and the embedded verb. In (17), for example, the woman that loves traveling *now* may very well not have loved it when she met Joseph. Things

get a bit more complex with verb complements. There, a present tense typically gives rise to the well-known ‘double-access’ reading (Smith, 1978). Given (18), for example, Mary’s love for John holds both at the time of utterance and at the time of John’s realization. Theories vary as to how exactly the double-access reading comes about, but it is standardly assumed that the semantic contribution of the present tense is to indicate the time of utterance (and only that). The simultaneity with the past matrix event comes from separate, possibly pragmatic mechanisms.⁸ If the primary indexical reading of the present is deemed implausible by context or world knowledge, as is the case in (19), we end up with a sentence that is infelicitous.

- (19) #Two thousand years ago, Joseph **found** out that Mary **loves** him.

(Ogihara and Sharvit 2012: (14))

This is sufficient to conclude that in English, the present embedded under past does not have a shifted reading. Instead it is interpreted the same as in simple root clauses, and it should not come as a surprise that, just like in (6), an embedded present in English is incompatible with *then*:

- (20) *A month ago, John **found** out that Mary **loves** him **then**.
 (21) *In his childhood, Joseph **met** a woman who **loves** traveling **then**.

Unlike English, however, languages such as Modern Greek, Modern Hebrew and Russian have a shiftable present. Recall from (11), repeated below as (22) with the addition of (23) and (24), that a present embedded under a past-tensed speech/attitude verb can be shifted, giving rise to a simultaneous reading. Contrary to English (see (18) and (19)), a shifted present does not force the event to hold at the time of utterance. Interestingly, however, the shifted present is still incompatible with ‘*then*’ in each of these languages.

- (22) To 2000, o Yanis **iksere** oti i Maria **ine** egkios (***tote**).
 The 2000, _{DET} Yanis **know.PAST** that the Maria **be.PRES** pregnant **then**
 ‘In 2000, Yanis knew that Maria was (LIT. is) pregnant then.’

Modern Greek (Tsilia 2021: (78))

- (23) Lifney alpayim šana, Yosef **xašav** še Miriam **ohevet** oto (***az**).
 Before 2,000 year, Yosef **think.PAST** that Miriam **love.PRES** him **then**
 ‘2,000 years ago, Yosef thought that Miriam loved (LIT: loves) him then.’

⁸For instance, Abusch (1988, 1997) proposed a pragmatic ‘Upper Limit Constraint’ that prohibits embedded tenses from being interpreted relative to any time later than the attitude holder’s ‘now’; it is through this constraint that the double-access reading is obtained, although a full discussion of this lies outside the scope of the paper. For alternative approaches, see e.g. Ogihara (1995, 1996); Altshuler and Schwarzschild (2013a); Bary and Altshuler (2015); Klecha (2018). Most of these approaches subscribe to the view that the English present embedded under past is anchored to the time of utterance, with Altshuler and Schwarzschild (2013a) arguing that the English present is both deictic and relative. See also Kusily (2020, 2024) who shows that simultaneous readings can marginally be obtained in English VP-fronting constructions:

- (1) Think that Mary is angry, John did.

(Kusily 2020: (8))

We put these constructions aside for the purposes of this paper, treating the English present as always referring to the time of the utterance. That being said, it would be interesting to see if the ‘*then*’-present puzzle holds in English VP-fronting constructions.

Modern Hebrew (Ogihara and Sharvit 2012: fn. 6)⁹

- (24) V dvuxtysjačnom godu Ivan **znal**, čto (***togda**) Maša **Ø** beremenna.
 In 2000 year Ivan **know.PAST** that **then** Maša **be.PRES** pregnant
 ‘In 2000, Ivan knew that Masha was (LIT: is) pregnant then.’

Russian (Tsilia 2021: (71), replicating Vostrikova 2019)

On the other hand, Modern Greek, Modern Hebrew and Russian are like English in *not* allowing tense shift in **relative clauses**. The embedded present in (25)–(27) unambiguously refers to a time that overlaps with the time of utterance like in (17); and like (21), it is incompatible with ‘*then*’.

- (25) Prin 20 chronia o Pavlos **sinerghastike** me enan andra pu **ine** proedros (***tote**).
 Before 20 years the Pavlos **collaborate.PFV.PAST** with a man who **be.PRES** president **then**.
 Only: ‘20 years ago, Pavlos collaborated with a man who is president (now).’

Modern Greek (adapted from Tsilia 2021: (32))

- (26) be-yalduto, **pagaš** Yosef iša še **ohevet** letayel (***az**).
 in-childhood-his **meet.PAST** Yosef woman COMP **love.PRES** traveling **then**
 Only: ‘In his childhood, Yosef met a woman who loves traveling (now).’

Modern Hebrew (adapted from Ogihara and Sharvit 2012: (20a))

- (27) Maša **videla** čeloveka, kotoryi (***togda**) **plačet**.
 Maša **see.PAST.IMP** man who **then** **cry.PRES**
 Only: ‘Maša saw a man who is crying (now).’

Russian (adapted from Kusumoto 1999: (129a))

Finally, Japanese differs from all the languages above in allowing the present to shift not only in the complement of speech/attitude verbs as in (28), but also in relative clauses as in (29) (see also Ogihara, 1996; Ogihara and Sharvit, 2012). In both environments the ‘*then*’-present puzzle arises.

- (28) ninenn mae, Yusuke-wa Yukiko-ga (***tooji**) ninnshinn **shite-iru** to **shitteita**.
 2-years before Yusuke-TOP Yukiko-nom **then** pregnant **be-PRES** with **know.PAST**.
 ‘2 years ago, Yusuke knew that Yukiko was (LIT: is) pregnant.’ Japanese

- (29) ninenn mae, Yusuke-wa (***tooji**) seifu de hatarai-te-**iru** hito to renkei o
 2-year before Yusuke-TOP **then** government LOC work.PROG.**PRES** person with collaboration OBJ
 hakat-te-**ita**.
 plan.IMP.FV.**PAST**.
 ‘Two years ago, Yusuke collaborated with a man who was (LIT: is) president.’ Japanese

The languages under consideration can therefore be divided into three subgroups: (i) English, where the present does not shift under past, whether in the complement of a speech/attitude verb or in relative clauses,

⁹Ogihara and Sharvit (2012) noted that the ‘*then*’-present configuration is unacceptable to ‘many, but admittedly not all’ native speakers. We will not attempt to capture this variation.

(ii) Modern Greek, Modern Hebrew and Russian, where the present can shift in speech/attitude reports but not in relative clauses, and (iii) Japanese, where the present can shift in both environments. The ‘*then*’-present puzzle arises consistently in all embedded constructions. The findings are summarized below in Table 1.

LANGUAGE	ENVIRONMENT	shifted present?	‘ <i>then</i> ’-present
Modern Greek	attitude report	✓	incompatible
	relative clause	✗	incompatible
Modern Hebrew	attitude report	✓	incompatible
	relative clause	✗	incompatible
Russian	attitude report	✓	incompatible
	relative clause	✗	incompatible
Japanese	attitude report	✓	incompatible
	relative clause	✓	incompatible
English	attitude report	✗	incompatible
	relative clause	✗	incompatible

Table 1: PRESENT UNDER PAST

2.2 Present under future

When the matrix verb is in the future tense, languages behave more uniformly. Almost all surveyed languages allow for a shifted present under future, both in speech/attitude reports and in relative clauses. With the exception of the English present, the puzzle arises with all other presents shifted under future we surveyed.¹⁰

(30) Present under future: attitude reports

a. Modern Greek

To 2030, i Maria **tha** pi oti **ine** sti filaki (***tote**).

In 2030, the Maria **FUT** say that **be.PRES** to-the jail **then**

‘In 2030, Maria will say that she is in jail (in 2030).’

(Tsilia 2021: (79))

b. Modern Hebrew

Be 2030, John **yagid** še hu Ø ba kele (***az**)

In 2030, John **say.FUT** that he **be.PRES** in jail **then**

‘In 2030, John will say that he is in jail (in 2030).’

(Tsilia 2021: (76))

¹⁰For Russian, further empirical investigation is needed. Tsilia (2021) has four consultants and reports that a simultaneous reading of present-under-future in relative clauses is possible as indicated in (31c). However, von Stechow and Grønn (2013) take the present in relative clauses to be necessarily deictic; see, e.g., ex.36 in von Stechow and Grønn (2013)). von Stechow and Grønn (2013) assume that Russian does not have a relative present, and analyze *budet* ‘will’ as an extensional quantifier (contrary to what we assume in this paper, where we treat it on a par with English *will*). More research is needed to establish whether the present in Russian relative clauses can shift under *budet* or not. We would also like to point out the possibility that there is within-speaker variation, especially given the big variation that is attested in the Russian tense system (see, e.g., Tsilia, 2021, Section 3.1 on variation with respect to whether past-under-past has a back-shifted or a simultaneous reading, and see also fn.30 for variation with respect to whether Russian ‘now’ is shiftable). For the purposes of this paper, we assume that a simultaneous reading of present-under-future in relative clauses is possible as shown in (31c).

c. Russian

V 2030, Ivan **skazet**, čto on (***togda**) **sidit** v tjur'me.
 In 2030, Ivan **say.FUT**, that he **then sit.IMPV.PRES** in jail
 'In 2030, Ivan will say that he is in jail (in 2030).' (Tsilia 2021: (72))

d. English

In 2030, John will say that he is in jail (**%then**). (Tsilia 2021: (85))

(31) Present under future: relative clauses

a. Modern Greek

To 2030, i Zoi **tha** ine pantremeni me kapon pu **ine** (***tote**) sti filaki.
 In 2030, the Zoi **FUT** be married to someone who **be.PRES then** to-the jail
 'In 2030, Zoe will be married to someone who is in jail.' (Tsilia 2021: (80))

b. Modern Hebrew

Be 2030, John **yihie** nasui le miši še Ø ba kele (***az**).
 In 2030, John **be.FUT** married to someone.FEM that **be.PRES** in jail **then**
 'In 2030, John will be married to someone who is in jail.' (Tsilia 2021: (77))

c. Russian

V 2030 godu, Ivan budet ženat na tom, kto (***togda**) **sidit** v
 In 2030 year, Ivan **be.FUT** marry.PFV.PASS to someone, who **then sit.IMPV.PRES** in
 tjur'me.
 jail
 'In 2030, Ivan will be married to someone who is in jail.' (Tsilia 2021: (73))

d. English

In 2030, John will be married to someone who is in jail (**then**).
 (Tsilia 2021: (87), adapted from von Fintel and Heim (1997-2021): §7, (4))

The patterns are summarized in Table 2. Two comments are in order. First, *tooji* 'then' in Japanese is only used with past expressions, so it is ruled out even with a matrix future tense as shown in (32). This is why we did not include Japanese examples in (30) and (31).

(32) *Two friends are making plans to go to the movies on Saturday. One of them says before they leave:*

{Sonotoki, ***Tooji**} ai-mashou.
 {That-time, ***Then**} see-let's
 'See you then.'

Secondly, Modern Greek, Modern Hebrew, and Russian—all languages with a shiftable present under past—exhibit the puzzle under future. However, in English, a language where the present can only shift under future, *then* is compatible with the present under future both in attitude reports as in (30d) (at least for

some speakers), as well as in relative clauses as in (31d). We will come back to this issue in Sections 5.3 and 6.

LANGUAGE	ENVIRONMENT	shifted present?	「 <i>then</i> 」-present
Modern Greek	attitude report	✓	incompatible
	relative clause	✓	incompatible
Modern Hebrew	attitude report	✓	incompatible
	relative clause	✓	incompatible
Russian	attitude report	✓	incompatible
	relative clause	✓	incompatible
Japanese	attitude report	✓	N/A
	relative clause	✓	N/A
English	attitude report	✓	variation (%)
	relative clause	✓	compatible

Table 2: PRESENT UNDER FUTURE

2.3 An interpretation *parameter* for tenses and 「*then*」

Let us now return to the earlier argument that the temporal anchors for tenses and 「*then*」 should be treated as an interpretation parameter rather than a (covert) temporal pronoun. The relevant LFs, (13) and (16), are repeated below as (33) and (34). Here we have also included the assignment function g in the array of interpretation parameters.

- (33) Yanis knew that Maria is pregnant (Lit. from Modern Greek)

$$\llbracket \text{Yanis know-}\mathbf{PAST}_{\substack{1, \\ \text{REF}}}^{\text{ANCHOR } 0} [\dots \text{Maria be-}\mathbf{PRES}_{\substack{2, \\ \text{REF}}}^{\text{ANCHOR } 1} \text{pregnant}] \rrbracket \dots g \dots$$
 (to be rejected)
- (34) Yanis knew that Maria is pregnant (Lit. from Modern Greek)

$$\llbracket \text{Yanis know-}\mathbf{PAST}_{\substack{1, \\ \text{REF}}} [\mathbf{OP}_{\pi} \text{Maria be-}\mathbf{PRES}_{\substack{2, \\ \text{REF}}} \text{pregnant}] \rrbracket \dots \pi^{\text{ANCHOR}}, g \dots$$

Both LFs result in the shifted reading where the temporal reference of the matrix tense $g(1)$ overlaps with that of the embedded present $g(2)$ (here and henceforth, $g(1) \circ g(2)$). The main difference between them is the treatment of the anchor, and accordingly the exact derivation of the simultaneous reading. In (33), the anchor is a (covert) pronoun, represented as the second index on the tense morpheme. The shifted reading is derived through the pronominal binding between the reference of the matrix past and the anchor of the shifted present (notice the co-indexation). In (34), on the other hand, tense shift is achieved through the shift of the temporal perspective π ; the latter is responsible for the interpretation of the embedded present, and is shifted through a propositional operator $\mathbf{OP}_{\pi}$. The details of how $\mathbf{OP}_{\pi}$ works will be introduced in Section 4.

Now consider how the incompatibility between the shifted present and 「*then*」 could be accounted for in either approach. We assume that 「*then*」 is similar to a tense in being sensitive to the temporal anchor, but, contrary to the present, it requires its reference to be *disjoint* from the anchor. The LFs of the ungrammatical 「*then*」-present sentences for each approach are in (35) below:

(35) *Yanis knew that Maria is pregnant then.

- a. $\llbracket \text{Yanis know-PAST}_{1,0} [\dots \text{Maria be-PRES}_{2,1} \text{ pregnant then}_{3,1}] \rrbracket \dots g \dots$ (pronominal anchor)
 b. $\llbracket \text{Yanis know-PAST}_{1,0} [\text{OP}_{\pi} \text{ Maria be-PRES}_2 \text{ pregnant then}_3] \rrbracket \dots \pi, g \dots$ (parametric anchor)

In both LFs, the embedded present and $\lceil \text{then} \rceil$ share the same temporal anchor valued at $g(1)$, although through different mechanisms (see above). And while the present requires its reference to overlap with the temporal anchor (i.e., $g(2) \circ g(1)$), $\lceil \text{then} \rceil$ requires its reference to be disjoint from it (i.e., $\neg g(3) \circ g(1)$). The conflicting conditions thus preclude $\lceil \text{then} \rceil$ from restricting the reference of the present (i.e. $g(2) \subseteq g(3)$ cannot hold), thus explaining their incompatibility.

The advantage of (35b) is that it is *guaranteed* that the present and $\lceil \text{then} \rceil$ share the same anchor, since OP_{π} is a propositional operator shifting the co-clausal present and $\lceil \text{then} \rceil$ simultaneously. It always yields a semantic contradiction, thus explaining the ungrammaticality of sentences like (35) across languages. On the other hand, in (35a), the sharing of anchors between the present and $\lceil \text{then} \rceil$ is largely *stipulated*. To see this, let's start with the assumption that anchors are indeed free variables that can indexed freely. This leads to potential over-generation, exemplified in (36):

(36) Yanis know-PAST_{1,0} [. . . Maria be-PRES_{2,1} pregnant then_{3,0}]

Here, although the shifted **PRES**_{2,1} is still anchored to the reference of the matrix past, $\lceil \text{then} \rceil$ shares its anchor $g(0)$ with the matrix past. This yields a potentially coherent reading: the reference of $\lceil \text{then} \rceil$ $g(3)$ could be a temporal interval that contains the reference of the shifted present, $g(2)$, as their anchors are disjoint ($g(1) < g(0)$). In other words, the pronominal analysis of anchors as such predicts that tense shift can *feed* the co-occurrence of the shifted present with $\lceil \text{then} \rceil$, contrary to the generalization.

One possible response to this argument could be that $\lceil \text{then} \rceil$ is not sensitive to a temporal anchor to begin with. That is, (35) may correspond to an LF where $\lceil \text{then} \rceil$ is only indexed with its reference:

(37) $\llbracket \text{Yanis know-PAST}_{1,0} [\dots \text{Maria be-PRES}_{2,1} \text{ pregnant then}_3] \rrbracket \dots g \dots$

But this does not get us anywhere. If $\lceil \text{then} \rceil$ is not anchored, then in order to account for the puzzle, there have to be direct restrictions on its reference $g(3)$ that prohibit it from overlapping with $g(2)$. Apart from the fact that such restrictions would be *ad hoc*, we would also predict that anything with the reference $g(2)$ cannot combine with then_3 either. This is not the case. Many Modern Hebrew and Russian speakers accept the co-occurrence of $\lceil \text{then} \rceil$ with an embedded *past*, which generates the same simultaneous reading (therefore the same temporal reference) as the shifted present:

- (38) lifney alpayim šana, Yosef **xašav** še Miriam **ahava** oto **az**.
 before two-thousand year Yosef **think.PAST** that Miriam **love.PAST** him **then**
 'Two thousand years ago Yosef believed that Miriam loved him then.'

Modern Hebrew (Ogihara and Sharvit 2012: (7a))

- (39) V 2016 godu Tanja **skazala**, čto **togda** Putin **byl** prezidentom Rossii.
 in 2016 year Tanja **say.PAST** that **then** Putin **be.PAST** president.INST Russia.GEN.
 ‘In 2016 Tanja said that Putin was the president of Russia then.’ Russian (Vostrikova 2019: (2))

Alternatively, one may attempt to augment the double-indexed pronominal analysis in (35b) with additional syntactic constraints that force the same anchor on the embedded tense and ‘*then*’ (e.g. see the *local binding* constraint in Anand 2007), thus avoiding over-generation and ruling out (36). This is a viable option that merits further empirical investigation, but here we point out a practical concern. If co-indexing is forced between the anchors of present and ‘*then*’, one should expect the same for their references, given the parallel treatment as temporal pronouns. However, as we can see in (40), multiple occurrences of the same tense morpheme may correspond to different temporal references.

- (40) a. Yesterday, John **found out** that Mary **was** drunk the day before.
 ~→ John found out yesterday that Mary was drunk two days ago
 (cf. the simultaneous reading in (14))
 b. Yesterday, John **found out** that Mary **was** drunk the day before and **was** feeling sick.
 ~→ John found out yesterday that Erica was drunk two days ago and was feeling sick yesterday.

In sum, to account for the ‘*then*’-present puzzle, clausemate tenses and ‘*then*’ should share the same temporal anchor, so that, when they shift, they shift together. This favors a parameter-based approach to the temporal anchor over a pronominal one. The next Section considers yet another existing analysis to tense shift that treats shifted tense on a par with deleted tense, and argues that the compatibility between deleted tenses and ‘*then*’ constitutes evidence against equating the shifted present with deleted tense. After that, we will return to the formal implementation of the parameter-based analysis that offers a solution to the ‘*then*’-present puzzle along with an account of the tense-shift typology.

3 Deleted tense and ‘*then*’

As mentioned in Section 1, another existing analysis for the shifted present uses a deletion mechanism that reduces the shifted present to a bound temporal pronoun, as sketched in (41):

- (41) Yanis knew that Maria is pregnant. (Lit. from Modern Greek)
 [[Yanis know-**PAST**₁ [. . . Maria be-**PRES**₁ pregnant]]] . . . π, g . . . (to be rejected)

Note that we have included the parameter π to be consistent with the previous discussion. Nevertheless, under the deletion-based analysis, the perspective (or the temporal anchor in general) does not play a role in the interpretation of the embedded tense. Instead, as suggested by the strike-through on **PRES**₁, the tense feature of the embedded tense is not interpreted at all, and its reference is simply bound by that of the higher tense **PAST**₁.

This deletion-based account, akin to what has been proposed in e.g. Ogihara and Sharvit (2012), essentially identifies the shifted present with *deleted past*. The latter also generates a simultaneous reading in past-under-past constructions such as (42). The deletion mechanism is often implemented as a sequence-of-tense (SoT) rule, formulated informally in (43).¹¹

- (42) John **found out** that Mary **was** drunk.
 [[John find-out-**PAST**₁ [. . . Maria be-**PAST**₁ drunk]]] . . . π, g . . .
 $\rightsquigarrow$ Mary was drunk at the time when John found out.

- (43) Sequence-of-tense (SoT)

The tense feature of a tense c-commanded by an agreeing tense may be deleted at the Logical Form.

Suspensions may already arise at this point: if SoT only applies to *agreeing* tenses, how can the deletion of the present feature be licensed under a past, as in (41)? But for now let us set this issue aside (see also fn. 11 for a more compatible implementation). The point we make is that it is empirically inadequate to collapse the *interpretations* of a shifted present and an embedded past that generates a simultaneous reading. Indeed, their difference becomes obvious when we look at their co-occurrence with ‘*then*’. While the shifted present is incompatible with ‘*then*’, an embedded past can co-occur with it, as indicated below:

- (44) V 2016 godu Tanja **skazala**, čto **togda** Putin **byl** prezidentom Rossii.
 in 2016 year Tanja **say.PAST** that **then** Putin **be.PAST** president.INST Russia.GEN.
 ‘In 2016 Tanja said that Putin was the president of Russia then.’

Russian (Vostrikova 2019: (2))

Still, at the moment, we cannot yet rule out that the shifted present behaves like a deleted past, since (42) is not necessarily analyzed as a deleted past.

For instance, both the matrix and the embedded past in (42) may be genuinely interpreted as relative to the time of utterance, and the simultaneous reading would thus simply arise as a result of co-reference (setting aside potential constraints such as ‘Non-Redundancy’ in Schlenker 2005). This corresponds to the LF in (45), with $g(1) = g(2)$.

- (45) John **found out** that Mary **was** drunk.
 [[John find-out-**PAST**₁ [Maria be-**PAST**₂ drunk]]] . . . π, g . . .

Such co-reference can be viewed as one of the results that follow from the ‘temporal *de re*’ mechanism à la Abusch (1997), and in fact, Vostrikova (2019) argued that the mechanism is what enables the co-occurrence

¹¹ An alternative implementation, advocated independently by Kratzer (1998) and von Stechow (2003), treats the embedded tense that generates the simultaneous reading as an inborn ‘zero’ tense, i.e., a tense morpheme that does not carry semantic tense features. As a reviewer points out, this enables tense deletion without surface-level morphological agreement between the higher and the embedded tense (similarly to (15)). Zero-tense gets spelled out as an agreeing past tense under a matrix past in SoT languages, and is realized as the default present tense in non-SoT languages. Since it is not necessary to assume an inborn zero tense for any language in our sample, the choice between an SoT rule that requires morphological identity and one that posits a zero tense will not be crucial for our purposes.

of past and ‘*then*’ in (44). Space precludes us from discussing the *de re* mechanism further (though see fn. 12 & 20).¹²

The point here is that in order to evaluate the deletion-based analysis for the shifted present, we need a more elaborate example that helps us control for coreference between the matrix and the embedded past. Fortunately, Abusch provides exactly that, with the sentence in (46):¹³

- (46) A week ago, John said that in ten days he would say to his girlfriend that they **were** meeting for the last time.

(46) has the reading that John said a week ago that three days from now, he will say to his girlfriend: “we *are* now meeting for the last time”. That is, John’s speech occurs at the same time as the meeting, three days from now. In this case, the most deeply embedded past tense (in bold) cannot be interpreted (*de re* or otherwise) as past relative to the time of utterance or relative to any other temporal reference established in the sentence, and therefore *has to be* deleted.

The same phenomenon is observed in Modern Greek (Sharvit 2018; Tsilia 2021), as shown in (47). Apart from English and Modern Greek, none of the other languages we consider (Modern Hebrew, Japanese and Russian) has a deleted tense. The counterparts of (46) in these languages lack the simultaneous reading on the lowest past tense.

- (47) Prin mia evdhomadha, o Jorghos ipe oti se dheka meres tha eleghe stin
 Before one week the Jorghos say.PAST that in ten days will say.IMP.FV.PAST to-the
 kopela tu oti **sinadjiondusan** ja teleftea fora.
 girlfriend of-his that meet.IMP.FV.**PAST** for last time
 ‘A week ago, Jorghos said that in ten days he would say to his girlfriend that they were meeting for the last time’
 Modern Greek (Sharvit 2018: (40))

¹²Vostrikova actually discusses both the ‘*then*’-present puzzle and data like (44), primarily in Russian. The analysis for the ‘*then*’-present puzzle is essentially a pronominal-based one as discussed in Section 2.3, with the additional assumption that the temporal anchors of embedded tense and ‘*then*’ are always bound by the embedding predicate. In other words, embedded tenses and ‘*then*’, if not read *de re*, always shift (we will argue in Section 7.2 that this is not cross-linguistically viable, due to the optionality of tense shift). This means that in order to deal with (44), apart from the *de re* mechanism that generates the simultaneous reading of the embedded past, something else is needed to ensure that *togda* ‘then’ does not shift—after all, a shifted ‘*then*’ cannot conceivably restrict the reference of a simultaneous past. Indeed, Vostrikova appeals to a *de re together* constraint, according to which tenses and temporal adverbials have to be interpreted *de re* together. But it is at least dubious whether this is an independently motivated constraint. For example, Romoli and Sudo (2009) showed that even within a single DP, its subconstituents need not be interpreted *de re* together. The possibility of a ‘mixed’ reading is demonstrated in the example below, where in the complex DP *the wife of the president*, *president* is read *de re*, whereas *wife* is read *de dicto*.

- (1) Context: Mary is watching television and sees Barack Obama, the actual president, and his sister besides him. Also, she doesn’t know who he is and she thinks that the woman besides him must be his wife.
 Mary thinks **the wife of the president** is nice. (Romoli and Sudo 2009: (20))

Relatedly, Vostrikova’s analysis cannot be straightforwardly extended to deleted tenses as in (46) and (47), which are incompatible with a *de re* analysis. Admittedly, Russian lacks deleted tense, but it has been reported to be acceptable for at least some speakers, especially with the presence of *togda* ‘then’ (Tsilia 2021: (119)). The advantage of the current proposal is that it accounts for Vostrikova’s various observations quite naturally, while also accounting for the behavior of ‘*then*’ with deleted tense. For an extensive discussion, see Zhao 2025: §2.8.1.

¹³Such constructions were first formulated in Abusch (1988) for English after Kamp and Rohrer’s (1984) discussions on French.

Now to the key observation: unlike the shifted present, a genuinely deleted tense is compatible with the adverbial ‘then’:

- (48) A week ago, John said that in ten days he would say to his girlfriend that they **were** meeting **then** for the last time.
- (49) Prin mia evdhomadha, o Jorghos ipe oti se dheka meres tha eleghe stin
 Before one week the Jorghos say.PAST that in ten days will say.IMPFV.PAST to-the
 kopela tu oti **sinadjiondusan tote** ja teleftea fora.
 girlfriend of-his that **meet.IMPFV.PAST then** for last time
 ‘A week ago, Jorghos said that in ten days he would say to his girlfriend that they were meeting then
 for the last time’ Modern Greek

Therefore, unlike what has been previously assumed, tense deletion and tense shift should be treated separately (and LFs such as (41) should be set aside). In Section 6, we will show that this could be done straightforwardly—we will propose that tense shift is due to perspective shift induced by $\mathbf{OP}_\pi$, while deleted tenses are stripped of their perspective-sensitivity altogether.

4 Towards the analysis: the compositional fragment

Having motivated the parameter-based analysis of tense shift, we now develop a compositional fragment for our system. As mentioned in the Introduction, our implementation draws heavily on theories of indexical shift, especially Anand and Nevins (2004) and Anand (2006). The analysis of tense shift mirrors that of indexical shift (see also Schlenker (2003); Barry (2012) a.o. for similar analyses of tense). Both result from the shifting of the corresponding parameters, which in turn captures the SHIFT TOGETHER effect. In what follows, we briefly introduce the indexical shift formalism of Anand and Nevins, and then extend it to our theory of tense. Key differences between the two systems will be identified along the way.

4.1 Preliminaries: A theory of indexical shift

Indexical shift refers to the phenomena where indexical expressions such as *I, you, here, now, today, yesterday* receive shifted interpretations in environments like speech/attitude reports. Below is an example from Nez Perce:¹⁴

- (50) Mipx Beth hi-neek--e [pro] kuu-se-∅]?
 where.to Beth.NOM 3SUBJ-think-P-REM.PAST [1SG GO-IMPERF-PRES]
 ✓Where did Beth_i think I was going?
 ✓Where did Beth_i think she_i was going? Nez Perce (Deal 2020: (5))

¹⁴Languages that have been reported to allow indexical shift include e.g. Amharic (Schlenker, 1999), Zazaki (Anand and Nevins, 2004; Anand, 2006), Japanese (Sudo, 2012), Uyghur (Shklovsky and Sudo, 2014), Korean (Park, 2016), Nez Perce (Deal, 2020) etc.

(50) is ambiguous between two possible interpretations of the embedded first-person pronoun. Under the non-shifted reading, the first-person pronoun draws reference from the utterance context and refers to the speaker. Under the shifted reading, it draws reference from a shifted context that represents the reported attitude, and refers to the author of that context, i.e. the attitude holder *Beth*.¹⁵

Indexical shift exhibits the so-called **SHIFT TOGETHER** constraint. This constraint dictates that all shiftable indexicals of the same type within a minimal clausal domain must draw their references from the same context (see e.g. Anand 2006: (100)). This is illustrated by the Nez Perce example in (51), where the embedded clause contains two indexical pronouns:

- (51) Ne-’níc-em pee-∅-n-e ’in-haama-na, [’im-im] ciq’aamqal
 1SG-older.sister-ERG 3/3-say-P-REM.PAST 1SG-husband-ACC [2SG-GEN dog(ERG)
 hi-twehkey’k-∅-e [’iin-e].
 3SUBJ-chase-P-REM.PAST 1SG-ACC]
- a. ✓My sister_s told my husband_h that his_h dog chased her_s. (both shifted)
- b. ?My sister_s told my husband_h that your dog chased me. (both non-shifted)
- c. ✗My sister_s told my husband_h that your dog chased her_s. (only the second shifted)
- d. ✗My sister_s told my husband_h that his_h dog chased me. (only the first shifted)
- Nez Perce (Deal 2020: (30))

As indicated, only two out of the four possible readings are attested. The default interpretation (51a) is one where both indexical pronouns retrieve their references from the *reported* speech context, where the first-person pronoun *’iin-e* is interpreted as ‘my sister’ (the reported speaker) and the second-person pronoun *’im-im* as ‘my husband’ (the reported addressee). The other possible reading, though less preferred, is one where neither indexical shifts (51b), i.e. the first-person pronoun refers to the speaker and the second-person pronoun to the addressee. Crucially, it is impossible for only one of the two embedded indexicals to shift, be it the first-person pronoun as in (51c) or the second-person pronoun as in (51d).

Anand and Nevins’ analysis, which is a faithful reincarnation of Kaplan’s (1979) two-dimensional semantics, characterized the interpretation function as relative to a context parameter *c* and an evaluation index *i*, along with the function *g* for variable assignment:

$$(52) \llbracket \cdot \rrbracket^{c,i,g}$$

Following Anand, the context parameter *c* and the evaluation index *i* are treated as structurally ‘homologous’, both consisting of quadruples of a speaker/author, an addressee, a time and a world (53).¹⁶

¹⁵The example contains overt *wh*-extraction from the embedded clause to show that (50) involves genuine indexical shift within indirect discourse, since *wh*-extraction is impossible from a syntactically opaque direct quotation:

(1) *Who did Mary say, “I handed the bag to ___”?
 cf. Who did Mary say she handed the bag to ___? (Deal 2020: ex. 11)

¹⁶The formal parallel is mainly for the benefit of expository simplicity, particularly regarding the derivations of context-/perspective-

- (53) a. $c = \langle c_s, c_a, c_t, c_w \rangle$
 b. $i = \langle i_s, i_a, i_t, i_w \rangle$

Indexical expressions draw their references directly from the relevant components of c . For instance:

- (54) a. $\llbracket \mathbf{I} \rrbracket^{c,i,g} = c_s$
 b. $\llbracket \text{now} \rrbracket^{c,i,g} = c_t$

Speech/attitude predicates, whose complements are the usual habitat of shifting phenomena, are modeled as quantifiers over the indices used to evaluate the embedded clause. Take *say* as an example:

- (55) $\llbracket \text{say } \varphi \rrbracket^{c,i,g} = \lambda x. \forall i' \in \mathbf{SAY}(x)(i). \llbracket \varphi \rrbracket^{c,i',g}$
 where $i' \in \mathbf{SAY}(x)(i)$ just in case i' represents a speech context compatible with what x says in i .

The final ingredient of the analysis is the *context-shifting operator* $\mathbf{OP}_c$. It functions as a Kaplanian ‘monster’ that overwrites the context parameter with the evaluation index:

- (56) $\llbracket \mathbf{OP}_c \varphi \rrbracket^{c,i,g} = \llbracket \varphi \rrbracket^{i,i,g}$

Let us illustrate the fragment so far with a sample derivation of indexical shift, shown in (57). Two key steps are involved. First, the intensional predicate binds the index i' used to evaluate the reported speech (see (57i)). Second, $\mathbf{OP}_c$ overwrites c with i' (see (57ii)), yielding the interpretation where the embedded indexical *me* refers to the speaker of the reported speech i'_s , i.e. Mark (see (57iii)).

- (57) $\llbracket \text{Mark says } \mathbf{OP}_c \text{ Erica hates } \boxed{\text{me}} \rrbracket^{c,i,g} = 1$, iff
 i. $\forall i' \in \mathbf{SAY}(\mathbf{M})(i_w). \llbracket \mathbf{OP}_c \text{ Erica hates } \boxed{\text{me}} \rrbracket^{c,i',g} = 1$, iff
 ii. $\forall i' \in \mathbf{SAY}(\mathbf{M})(i_w). \llbracket \text{Erica is hates } \boxed{\text{me}} \rrbracket^{i',i',g} = 1$
 iii. $\forall i' \in \mathbf{SAY}(\mathbf{M})(i_w). \text{HATE}(\boxed{i'_s})(\mathbf{E})(i'_w)$

The operator-based analysis accounts for the SHIFT TOGETHER constraint by assuming that $\mathbf{OP}_c$ only applies to propositional nodes. This is illustrated in (58). Either $\mathbf{OP}_c$ applies to the whole (embedded) clause as in (58b) and both indexicals draw their references from a shifted parameter, or $\mathbf{OP}_c$ does *not* apply and both indexicals are interpreted relative to the context of the utterance. No intermediate options are available.

- (58) a. $\llbracket \text{your dog is chasing me} \rrbracket^{c,i,g} = \text{CHASE}(\text{DOG-OF}(\boxed{c_a})(\boxed{c_s}))(i_w)$
 b. $\llbracket \mathbf{OP}_c \text{ your dog is chasing me} \rrbracket^{c,i,g} = \llbracket \text{your dog is chasing me} \rrbracket^{i,i,g}$
 $= \text{CHASE}(\text{DOG-OF}(\boxed{i_a})(\boxed{i_s}))(i_w)$

shifting. Conceptually, the context and the index play very different roles in interpretations—the former represents the *circumstances* of the utterance, whereas the latter is used to evaluate the truth values of the propositional content of an utterance in specific contexts, as standardly assumed.

Comparing (58) with the discussion in Section 2, it is clear that the perspective shifting operator behaves very similarly. In what follows, we will essentially adopt the same strategy to derive tense shift on a par with indexical shift, and leverage SHIFT TOGETHER to ensure that clausemate tense and ‘*then*’ are always anchored to the same temporal perspective. The rest of the section will prepare for the formal analysis by developing a fragment that is able to handle compositions with tenses.

4.2 Preliminary: Compositions

The first modification we would like to make is on the interpretation function $\llbracket \cdot \rrbracket$, for reasons independent from our empirical concerns. Instead of taking both the context c and index i as parameters, we will ‘unload’ the index i into the object language, as shown in (59). The place holder ‘...’ in the array of parameters is reserved for the temporal perspective.

$$(59) \quad \llbracket \cdot \rrbracket^{c, \dots, g} = \dots \lambda i. \dots$$

This is motivated by the well-known problems of treating the index as a parameter (which Anand (2006): p. 108 already noted). Systems that treat the index as a parameter lack the necessary expressive power to capture a substantial fragment of natural language, unlike systems where evaluation indices are treated as variables (see e.g. Cresswell 1990). Allowing indices in the object language is also in better conceptual alignment with the pronominal treatment of tenses that we will adopt.¹⁷

The set of compositionally relevant types τ is defined in (60). As usual, each basic type corresponds to a domain of objects in the model— $\mathcal{D}_e$ is the domain of individuals, $\mathcal{D}_t = \{0, 1\}$, $\mathcal{D}_s$ the set of worlds, and $\mathcal{D}_i$ the domain of times, which we will take as the set of closed time intervals. Derivative types include types of functions $\langle \tau, \tau \rangle$ and products $\tau \times \tau$.

$$(60) \quad \tau ::= e \mid t \mid s \mid i \mid \langle \tau, \tau \rangle \mid \tau \times \tau$$

The product type $\tau \times \tau$ corresponds to ordered pairs of objects; and since either object may itself be a product type, it can be used to construct the types of tuples of any denumerable arity. In this paper, product types are solely reserved for indices and contexts, modeled as quadruples of speaker/author (e), addressee (e), time (i) and world (s). We will write κ in short for the corresponding composite type:

$$(61) \quad \kappa ::= e \times e \times i \times s$$

A list of relevant lexical entries is collected in Table 3. Names (e.g. Nate, Erica) and indexicals (e.g. I, me) are of type e , tenses and ‘*then*’ of type i . Common nouns like *person* have the type of intensional properties $\langle \kappa, \langle e, t \rangle \rangle$. Infinitives like *be angry* and *hate* are intensional predicates of the corresponding valencies, with an additional time argument, to be saturated by tense. Speech/attitude verbs like *say* have the

¹⁷The price to pay for this alteration is a slightly more complex compositional system that involves functional types corresponding to *intensional* properties and propositions. The system we adopt is essentially a variant of Gallin’s (1975) Ty2, supplemented with a few idiosyncratic composition rules adapted from more familiar sources (e.g. Heim and Kratzer 1998) for simpler exposition.

type $\langle \kappa, \langle i, \langle \langle \kappa, t \rangle, \langle e, t \rangle \rangle \rangle \rangle$. After combining with an index and a time argument representing the time of the reported speech, it relates the complement proposition (of type $\langle k, t \rangle$) with the speaker/attitude holder. Finally, quantifiers receive their usual extensional type $\langle \langle e, t \rangle, \langle \langle e, t \rangle, t \rangle \rangle$. The semantics of tenses will be revised in Section 4.3.

lexical items	types	$\llbracket \cdot \rrbracket^{c, \dots, g}$
I/me, Nate, Erica	e	$c_s, \mathbf{N}, \mathbf{E}$
person	$\langle \kappa, \langle e, t \rangle \rangle$	$\lambda i. \lambda x. \mathbf{PERSON}(i)(x)$
be angry	$\langle \kappa, \langle i, \langle e, t \rangle \rangle \rangle$	$\lambda i. \lambda t. \lambda x. \mathbf{ANGRY}(i)(t)(x)$
hate	$\langle \kappa, \langle i, \langle e, \langle e, t \rangle \rangle \rangle \rangle$	$\lambda i. \lambda t. \lambda x. \lambda y. \mathbf{HATE}(i)(t)(x)(y)$
say	$\langle \kappa, \langle i, \langle \langle \kappa, t \rangle, \langle e, t \rangle \rangle \rangle \rangle$	$\lambda i. \lambda t. \lambda p. \lambda x. \forall i' \in \mathbf{SAY}(i)(t)(x). p(i')$ (to be revised)
some	$\langle \langle e, t \rangle, \langle \langle e, t \rangle, t \rangle \rangle$	$\lambda P. \lambda Q. \exists x. P(x) \wedge Q(x)$
every	$\langle \langle e, t \rangle, \langle \langle e, t \rangle, t \rangle \rangle$	$\lambda P. \lambda Q. \forall x. P(x) \rightarrow Q(x)$
PRES_n, PAST_n	i	$g(n)$ (to be revised)

Table 3: list of lexical entries

The composition is built upon the basic functional application (δ and σ are meta-variables for types):

(62) FUNCTIONAL APPLICATION (FA)

If α is a binary branching node with two children $\{\beta, \gamma\}$, where β is of type $\langle \sigma, \delta \rangle$ and $\gamma : \sigma$ (i.e., γ is of type σ), then $\alpha : \delta$ and $\llbracket \alpha \rrbracket^{c, \dots, g} = \llbracket \beta \rrbracket^{c, \dots, g}(\llbracket \gamma \rrbracket^{c, \dots, g})$

We will also make use of *Intensional* Functional Application (IFA) to deal with intensional expressions with type- κ arguments. There are three possible ways of applying IFA: (i) applying an intensional function to a non-intensional argument as in (63.1), (ii) applying a non-intensional function to an intensional argument as in (63.2), and (iii) applying an intensional function to an intensional argument as in (63.3). All of them are simple extensions to FA, essentially percolating the index dependency from the child node(s) to the parent node. Notice that the parent node α is always of the type $\langle \kappa, \delta \rangle$.

(63) INTENSIONAL FUNCTIONAL APPLICATION (IFA)

If α is a binary branching node with two children $\{\beta, \gamma\}$:

1. if $\beta : \langle \kappa, \langle \sigma, \delta \rangle \rangle$ and $\gamma : \sigma$, then $\alpha : \langle \kappa, \delta \rangle$ and $\llbracket \alpha \rrbracket^{c, \dots, g} = \lambda i_{\kappa}. \llbracket \beta \rrbracket^{c, \dots, g}(i)(\llbracket \gamma \rrbracket^{c, \dots, g})$
2. if $\beta : \langle \sigma, \delta \rangle$ and $\gamma : \langle \kappa, \sigma \rangle$, then $\alpha : \langle \kappa, \delta \rangle$ and $\llbracket \alpha \rrbracket^{c, \dots, g} = \lambda i_{\kappa}. \llbracket \beta \rrbracket^{c, \dots, g}(\llbracket \gamma \rrbracket^{c, \dots, g}(i))$
3. if $\beta : \langle \kappa, \langle \sigma, \delta \rangle \rangle$ and $\gamma : \langle \kappa, \sigma \rangle$, then $\alpha : \langle \kappa, \delta \rangle$ and $\llbracket \alpha \rrbracket^{c, \dots, g} = \lambda i_{\kappa}. \llbracket \beta \rrbracket^{c, \dots, g}(i)(\llbracket \gamma \rrbracket^{c, \dots, g}(i))$

To deal with adjunction and quantification, the following versions of Predicate Modification and Predicate Abstraction rules are adopted. As usual, Predicate Modification composes two intensional properties over objects of a type σ (e.g. individuals ($\sigma = e$) or times ($\sigma = i$)), and returns an intensional property of the same type that conjoins them. Predicate Abstraction creates an intensional property from an intensional proposition by lambda-abstracting over a variable with the number index n . Note that both rules ensure the resulting intensional properties take an evaluation index as the first argument.

(64) PREDICATE MODIFICATION

If α is a binary branching node with two children $\{\beta, \gamma\}$, where $\beta : \langle \kappa, \langle \sigma, t \rangle \rangle$ and $\gamma : \langle \kappa, \langle \sigma, t \rangle \rangle$, then $\alpha : \langle \kappa, \langle \sigma, t \rangle \rangle$ and $\llbracket \alpha \rrbracket^{c, \dots, g} = \lambda i_{\kappa} \lambda x_{\sigma}. \llbracket \beta \rrbracket^{c, \dots, g}(i)(x) \wedge \llbracket \gamma \rrbracket^{c, \dots, g}(i)(x)$.

(65) PREDICATE ABSTRACTION

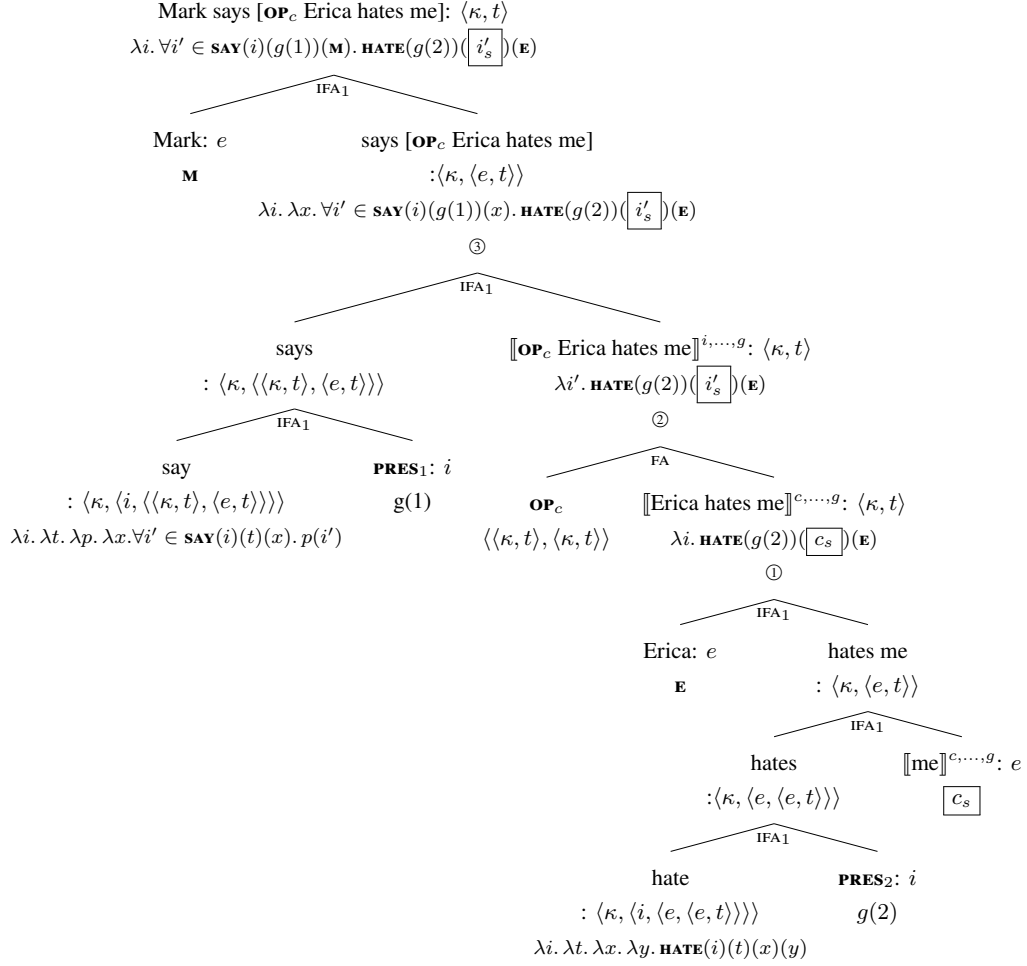
If α is a binary branching node with two children $\{\lambda_n, \beta\}$, where $\beta : \langle \kappa, t \rangle$, then $\alpha : \langle k, \langle \sigma, t \rangle \rangle$ and $\llbracket \alpha \rrbracket^{c, \dots, g} = \lambda i_k \lambda x_{\sigma}. \llbracket \beta \rrbracket^{c, \dots, g[n \mapsto x]}$

Let us now illustrate the fragment by working out the detailed derivations of indexical shift sketched in (57). The context-shifting operator $\mathbf{OP}_c$ is defined as follows:

$$(66) \quad \llbracket \mathbf{OP}_c \varphi \rrbracket^{c, \dots, g} = \lambda i_{\kappa}. \llbracket \varphi \rrbracket^{i, \dots, g}(i)$$

This is essentially the same as (56), where $\mathbf{OP}_c$ substitutes the context parameter with the index parameter. Once again, $\mathbf{OP}_c$ is assumed to only apply to propositional nodes (of type $\langle \kappa, t \rangle$). The derivation of (57) proceeds as in (67). The composition rule implemented at each step is specified under each branching node.

$$(67) \quad \llbracket \text{Mark says } \mathbf{OP}_c \text{ Erica hates } \boxed{\text{me}} \rrbracket^{c, \dots, g}$$



Note the key steps: ① is the result of computing the embedded proposition of type $\langle \kappa, t \rangle$, where the first-person indexical is shown to draw its reference from the (local) context; ② is the result of applying $\mathbf{OP}_c$ to ①, where the context parameter is overwritten by the abstracted index i' ; in ③ the speech predicate *say* binds the index, where the indexical corresponds to the attitude holder, Mark, rather than the speaker. The final result suggests that the sentence is true in c if and only if what Mark says at the time $g(1)$ overlapping with the time of utterance c_t is that Erica hates Mark. These are indeed the desired truth conditions.

4.3 Adding perspectives

As mentioned before, our theory of tenses mirrors that of indexicals in that tenses are interpreted relative to a temporal **perspective** parameter π , and tense shift is the result of perspectival shift. Specifically, the temporal perspective will be added into the list of interpretation parameters. The addition of π thus completes the

array of parameters of the interpretation function, as shown in (68) (cf. (59)). We will shortly justify the addition of the perspective parameter in Section 7.

$$(68) \quad \llbracket \cdot \rrbracket^{c,\pi,g} = \dots \lambda i. \dots$$

Unlike the context parameter c and evaluation index i , which are both modeled as quadruples of type κ , the perspective π is taken to be a simple time parameter (of type i) that denotes a time interval.¹⁸ Tenses and ‘*then*’ are treated as temporal pronouns with perspectival *presuppositions*, as shown in (69)–(71). Recall that ‘ $\circ$ ’ is read as ‘temporally overlaps with’, and ‘ $<$ ’ as ‘temporally precedes’.

$$(69) \quad \llbracket \mathbf{PRES}_n \rrbracket^{c,\pi,g} = g(n) \text{ only if } \boxed{g(n) \circ \pi}, \text{ otherwise undefined.}$$

$$(70) \quad \llbracket \mathbf{PAST}_n \rrbracket^{c,\pi,g} = g(n) \text{ only if } \boxed{g(n) < \pi}, \text{ otherwise undefined.}$$

$$(71) \quad \llbracket \mathbf{then}_n \rrbracket^{c,\pi,g} = g(n) \text{ only if } \boxed{\neg(g(n) \circ \pi)}, \text{ otherwise undefined.}$$

Other than the addition of π , everything is quite standard. $\mathbf{PRES}_n$ presupposes that its reference $g(n)$ overlaps with the temporal perspective π (i.e. their intersection is non-empty). $\mathbf{PAST}_n$ presupposes that $g(n)$ *precedes* π (i.e. all time points in the interval $g(n)$ precede all time points in π), and $\mathbf{then}_n$ is defined only if $g(n)$ is disjoint from π .¹⁹

Nothing significant hinges on the specific theory of presuppositions one assumes, so we can assume that they percolate along the derivations in a familiar fashion. For illustration, we will use the speech predicate *say*, treated as a presupposition filter, following Heim (1992):

$$(72) \quad \llbracket \mathbf{say} \rrbracket^{c,\pi,g} = \lambda i_{\kappa}. \lambda t. \lambda p_{\langle \kappa, t \rangle}. \lambda x. \forall i' \in \mathbf{SAY}(i)(t)(x). \llbracket \varphi \rrbracket^{c,\pi,g} \\ \text{only if } \forall i' \in \mathbf{SAY}(t)(x)(i_w), \llbracket \varphi \rrbracket^{c,\pi,g}(i') \text{ is defined, otherwise undefined}$$

With that, here is a revised list of lexical entries, upgrading Table 3 with the semantics of tenses above:

¹⁸This is mainly for reasons of simplicity, as we are only dealing with tense shift here. As mentioned in the Introduction, it remains to be explored whether tenses and the temporal perspective can be merged into a natural class of perspectives and perspective-sensitive items.

¹⁹For now we set aside the potential complication that the perspective-sensitivity of ‘*then*’ may be better characterized as an *anti-presupposition* (Percus 2006). This is shown in the following example (p.c. Philippe Schlenker):

- (1) When it’s sunny but cold, I don’t go out; when it’s sunny and warm, *then* I go out. So now I’m going out.

Then seems to refer to a time that at least includes the present, so the disjoint inference is not as stable as a presupposition might be. However, it seems that *then* here is restricted to the clause-initial position as in (1). For example, compare (1) and (2):

- (2) When it’s sunny and warm, I (??then) go out (??then).

This suggests that the argument may not hold for the clause-final (71), which is involved in most of our data points.

lexical items	types	$\llbracket \cdot \rrbracket^{c,\pi,g}$
I/me, Nate, Erica	e	$c_s, \mathbf{N}, \mathbf{E}$
person	$\langle \kappa, \langle e, t \rangle \rangle$	$\lambda i . \lambda x . \mathbf{PERSON}(i)(x)$
be angry	$\langle \kappa, \langle i, \langle e, t \rangle \rangle \rangle$	$\lambda i . \lambda t . \lambda x . \mathbf{ANGRY}(i)(t)(x)$
hate	$\langle \kappa, \langle i, \langle e, \langle e, t \rangle \rangle \rangle \rangle$	$\lambda i . \lambda t . \lambda x . \lambda y . \mathbf{HATE}(i)(t)(x)(y)$
say	$\langle \kappa, \langle i, \langle \langle \kappa, t \rangle, \langle e, t \rangle \rangle \rangle \rangle$	$\lambda i . \lambda t . \lambda p . \lambda x . \forall i' \in \mathbf{SAY}(i)(t)(x) . p(i')$ only if $\forall i' \in \mathbf{SAY}(t)(x)(i_w), \llbracket \varphi \rrbracket^{c,\pi,g}(i')$ is defined
some	$\langle \langle e, t \rangle, \langle \langle e, t \rangle, t \rangle \rangle$	$\lambda P . \lambda Q . \exists x . P(x) \wedge Q(x)$
every	$\langle \langle e, t \rangle, \langle \langle e, t \rangle, t \rangle \rangle$	$\lambda P . \lambda Q . \forall x . P(x) \rightarrow Q(x)$
PRES _n	i	$g(n)$ only if $g(n) \circ \pi$
PAST _n	i	$g(n)$ only if $g(n) < \pi$
then _n	i	$g(n)$ only if $\neg(g(n) \circ \pi)$

Table 4: list of lexical entries

Just like the context parameter c , the perspective parameter π may be shifted by an operator $\mathbf{OP}_\pi$ that applies only to propositional nodes of type $\langle \kappa, t \rangle$. The operator overwrites the perspective parameter of its prejacent with the time component of the evaluation index i_t :

$$(73) \quad \llbracket \mathbf{OP}_\pi \varphi \rrbracket^{c,\pi,g} = \lambda i_\kappa . \llbracket \varphi \rrbracket^{c,i_t,g}(i)$$

Finally, we have the following truth convention:

$$(74) \quad \text{TRUTH CONVENTION: A sentence } S \text{ with the LF } \varphi \text{ is true relative to the utterance context } c \text{ and assignment function } g \text{ iff } \llbracket \varphi \rrbracket^{c,c_t,g}(c) = 1.$$

What this truth convention essentially says is that evaluate the truth value of an utterance at a context c , the default value for the index is the context c , and the default value for the perspective is the temporal component of the context c_t . This will become relevant in relative clauses, where there is no intensional operator to shift the context and the perspective.

5 Solution to the ‘then’-present puzzle

We now have all the necessary pieces to solve the puzzle. As already mentioned, tense shift in our account is achieved through the $\mathbf{OP}_\pi$ operator, which shifts the temporal perspective π . As for the incompatibility of ‘then’ with the present, it follows from the conflicting presuppositions between them. More specifically, the present presupposes overlap with the perspective while ‘then’ presupposes disjointness from it. In this Section, we show how this exactly works compositionally, as well as how we derive the tense-shift typology.

5.1 Root clauses

Let us start with the simplest sentence we need to account for, namely a matrix present with ‘then’, as in (75):

All relevant lexical entries have been given in Table 4. But how does the temporal adjunct ‘*then*’ compose with the rest of the sentence? We assume that there is a silent preposition *during* before ‘*then*’. When combined with *then*₁ below, *during* gives an intensional temporal property corresponding to the set of time intervals contained within *then*₁’s reference.

- (76) a. $\llbracket \text{during} \rrbracket^{c, \pi, g} : \langle \kappa, \langle i, \langle i, t \rangle \rangle \rangle := \lambda i_{\kappa} . \lambda t_i . \lambda t'_i . t' \subseteq t$
 b. $\llbracket \text{then}_1 \rrbracket^{c, \pi, g} = g(1)$ only if $\neg(g(1) \circ \pi)$
 c. $\llbracket \text{during then}_1 \rrbracket^{c, \pi, g} = \lambda i . \lambda t . t \subseteq g(1)$ only if $\neg(g(1) \circ \pi)$

The tense is quantifier-raised, so *during then* composes with the un-tensed VP via PREDICATE MODIFICATION (see (64)). Then, this new temporal property takes the present as an argument, as the LF in (77) shows. Remember that this is the predicted LF of an ungrammatical sentence.

$\llbracket \text{Erica be-}\mathbf{PRES}_1 \text{ angry then}_2 \rrbracket^{c,\pi,g} : \langle \kappa, t \rangle$
 $\lambda i. \mathbf{ANGRY}(i)(g(1))(\mathbf{E}) \wedge \boxed{g(1) \subseteq g(2)}$
 only if $\boxed{g(1) \circ \pi}$ and $\boxed{\neg(g(2) \circ \pi)}$

IFA₁

$\llbracket \mathbf{PRES}_1 \rrbracket^{c,\pi,g} : i$
 $g(1) \text{ only if } \boxed{g(1) \circ \pi}$ $\lambda i . \lambda t. \mathbf{ANGRY}(i)(t)(\mathbf{E}) \wedge \boxed{t \subseteq g(2)} \text{ only if } \boxed{\neg(g(2) \circ \pi)}$

PM

$\langle \kappa, \langle i, t \rangle \rangle$
 $\lambda i . \lambda t. \mathbf{ANGRY}(i)(t)(\mathbf{E})$ $(\text{during}) \text{ then}_2 : \langle \kappa, \langle i, t \rangle \rangle$
 $\lambda i . \lambda t. \boxed{t \subseteq g(2)} \text{ only if } \boxed{\neg(g(2) \circ \pi)}$

PA **IFA₁**

$\lambda_1 \quad \text{Erica be-}t_1 \text{ angry: } \langle \kappa, t \rangle$
 $\lambda i. \mathbf{ANGRY}(i)(g(1))(\mathbf{E})$ $(\text{during}): \langle \kappa, \langle i, \langle i, t \rangle \rangle \rangle$ $\llbracket \text{then}_2 \rrbracket^{c,\pi,g} : i$

IFA₁ **IFA₁**

Erica: e $t_1 : i$
E $g(1)$

be angry
 $\langle \kappa, \langle i, \langle e, t \rangle \rangle \rangle$
 $\lambda i . \lambda t . \lambda x. \mathbf{ANGRY}(i)(t)(x)$

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of **PRES**₁, i.e. $g(1) \circ \pi$, (ii) the perspectival presupposition of **then**₂, i.e. $\neg(g(2) \circ \pi)$, and (iii) the temporal restriction $g(1) \subseteq g(2)$. These are conflicting conditions: if (i) and (ii) are satisfied, the assertion cannot hold. This is because the part of $g(1)$ that overlaps with π cannot be contained within $g(2)$, since the latter is disjoint from π . The ‘*then*’-present incompatibility thus follows as a semantic anomaly—there is no context in which a sentence like (75) is true with its presuppositions satisfied.

5.2 Shifted present in past speech/attitude reports

How about the incompatibility of ‘*then*’ with a shifted present? Recall that this was the case for Modern Greek, Modern Hebrew, Russian and Japanese, all languages with a shiftable present in speech/attitude reports. Using English pseudo-glossing, these constructions correspond to the following LF:

- (78) *Nate said Erica is angry then.

$$\llbracket \text{Nate say-PAST}_1 \llbracket \text{OP}_\pi \text{ Erica be-PRES}_2 \text{ angry then}_3 \rrbracket \rrbracket^{c, \pi, g}$$

The derivation for (78) is in (79). The tense in the embedded clause shifts similar to how indexicals shift in (67). There are two crucial steps: at step ①, OP_π applies, overwriting the temporal perspective with an abstract temporal index i'_t . Since the operator shifts the perspective of the whole embedded clause, **PRES**₂ and **THEN**₃ shift together; at step ②, the intensional predicate *say* binds the evaluation index i' of the embedded clause, associating the time component i'_t with the time of the reported speech. The embedded **PRES**₂ ends up denoting a time overlapping with the time of the attitude $g(1)$, while then_3 a time disjoint from it. Like before, a conflict arises between the perspectival presuppositions of **PRES**₂ and then_3 and the asserted temporal restriction $g(2) \subseteq g(3)$. This explains why ‘*then*’ cannot be used to restrict the shifted present.²⁰

²⁰The analysis proposed here predicts that a shifted present always gives rise to a temporal *de se* reading. This is due to the fact that OP_π overwrites π with the temporal index i_t that is understood as the time of the subject attitude holder’s perceived ‘now’ at the time of the attitude. So our analysis of a sentence with shifted present such as *Nate said Erica is angry* (LIT. in Modern Greek, Modern Hebrew, Russian, Japanese) should be understood as reporting a past speech of Nate that says “Erica is angry now”. But in principle, just like individual pronouns may receive *de re* (and non-*de se*) interpretations in attitude reports, one might expect embedded tenses to do so as well. Concretely, a *de re* interpretation of a shifted present is necessary if it can be used in a speech/attitude report where the embedded event(uality) overlaps with the actual attitude time, but not according to the attitude holder, as in the following scenario:

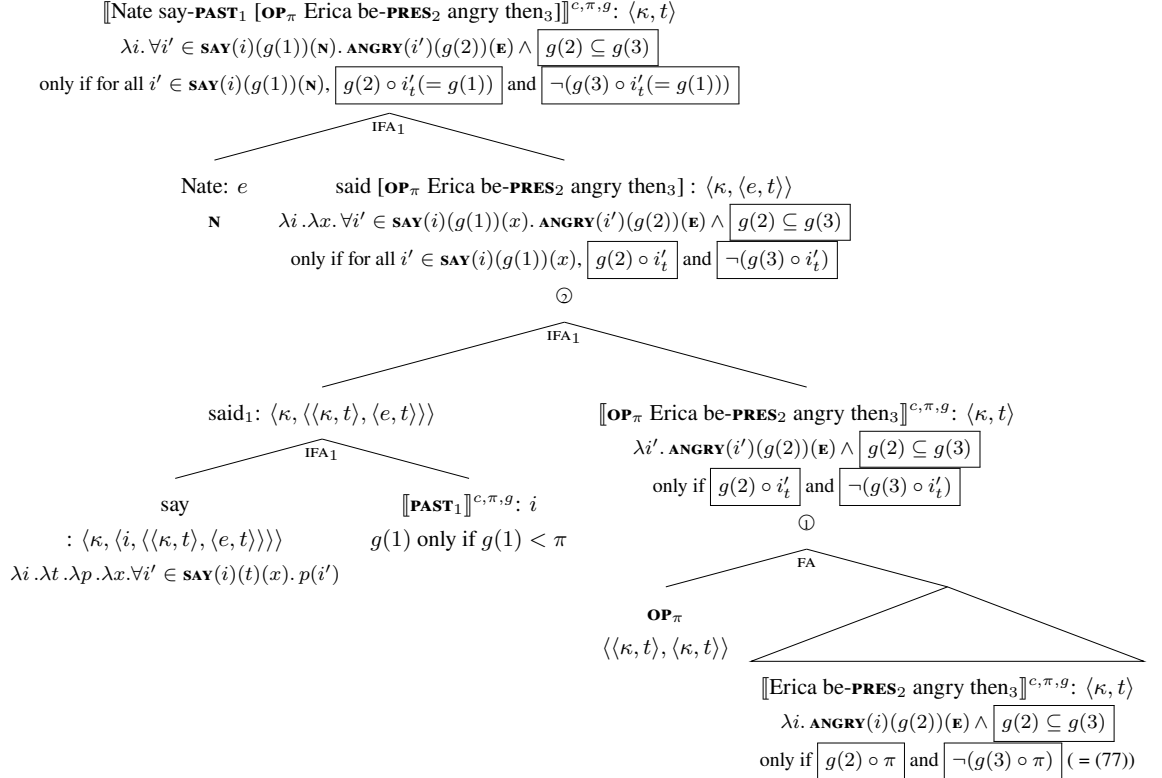
- (1) Context: Alex knows that Bambi has a road test on Oct 2. On that day, however, Alex mistakenly believed that it was Oct 3, and she told her friend Chris: “Bambi had the road test yesterday”. Realizing that Alex was mistaken about the date, Chris reports several days later. . .

%On October 2, Alex told me that Bambi **has** a driving test.

(LIT. in Modern Greek)

If the above utterance with an embedded present is acceptable in languages that allow for tense shift (we found one Modern Greek speaker who accepts it and one who rejects it, so more research is needed), then our analysis needs to be enriched to cover these *de re* readings of the shifted present. We leave a careful investigation of this empirical question for future research. But we would like to point out that the operator-based analysis can be made compatible with the temporal *de re* reading. In fact, Deal (2020: §3.2.3) offered a concrete extension to the operator-based analysis of indexical shift that captures cases where shifted indexicals receive non-*de se* readings (as observed in languages such as Zazaki and Uyghur). There, the key idea is that instead of overwriting the (shifted) context with the index bound by the attitude predicate, the shifting operator might optionally overwrite the context with the relevant coordinate of the attitude in the actual world. Accordingly, in the case of tense shift, we can assume that OP_π may overwrite the temporal perspective with the time that the attitude actually occurs.

(79) $\llbracket \text{Nate say-PAST}_1 [\text{OP}_\pi \text{ Erica be-PRES}_2 \text{ angry then}_3] \rrbracket^{c,\pi,g}$



Note again the similarity with indexical shift. Steps ① and ② go hand in hand: for the perspective shift to take effect, an intensional operator such as an attitude verb is required to bind the abstract index i' introduced by OP_π . Otherwise, the dependency on i' would percolate all the way to the top, and the TRUTH CONVENTION in (74) would reset the perspective to its default value, i.e. the time of the utterance; the result would be that embedded tenses and ‘then’ would still receive non-shifted readings. For languages such as Modern Greek, Modern Hebrew and Russian, this feature of our theory makes the correct prediction that tense shift is unavailable in relative clauses under past, as they are extensional environments with no intensional operator to bind the index.²¹ There is only one exception to this, as we saw in Section 2, namely Japanese. In Japanese, surprisingly, the present can also shift in relative clauses, despite the absence of an intensional operator. This is a puzzle that is entirely independent from ours, and thus we will not attempt to solve it. However, there is a way of technically handling these cases in our framework, namely by assuming that Japanese tenses can bind indices, like intensional verbs. We derive the relevant examples under this assumption in Appendix A.

So far, we have shown how ‘then’ and the present have conflicting presuppositions, whenever the present

²¹Recall that this does not hold for relative clauses under the future (Section 2.2). The next subsection will resolve this tension by treating the future itself as an intensional operator, following Ogiwara and Sharvit (2012) among others.

shifts. Another environment where the present shifts is under a future modal. We explore this next.

5.3 Shifted present under future

As we saw in Section 2, all languages surveyed in this paper allow for the present to shift under future, both in attitude reports as well as in relative clauses. We suggest, following Ogihara and Sharvit (2012), that the reason for the indiscriminate licensing condition is that the future itself contains an intensional operator that quantifies over evaluation indices.

Concretely, the semantics of $\ulcorner will \urcorner$ is decomposed into an index-binding operator **WOLL** and a present tense, as shown in (80). (Again, the corner quote $\ulcorner \cdot \urcorner$ indicates a placeholder for all future operators.) The modal component of **WOLL** is represented as the (universal) quantification over a set of indices i' generated by the abstract ‘accessibility’ function **ACC** given the evaluation index i at time t . **WOLL** also contains a future component that requires the time coordinate i'_t for each i' to temporally follow the time argument t . The untensed preajacent p is then evaluated across all accessible indices i' at i'_t .²²

$$\begin{aligned}
 (80) \quad \llbracket will_n \rrbracket^{c,\pi,g} &= \llbracket \mathbf{WOLL-PRES}_n \rrbracket^{c,\pi,g} : \langle \kappa, \langle \langle \kappa, \langle i, t \rangle \rangle, t \rangle \rangle, \text{ where} \\
 \llbracket \mathbf{WOLL} \rrbracket^{c,\pi,g} &: \langle \kappa, \langle i, \langle \langle \kappa, \langle i, t \rangle \rangle, t \rangle \rangle \rangle \\
 &:= \lambda i . \lambda t . \lambda p_{\langle \kappa, \langle i, t \rangle \rangle} . \forall i' \in \mathbf{ACC}(i)(t) . i'_t > t \wedge \llbracket p \rrbracket^{c,\pi,g}(i')(i'_t) \\
 &\quad \text{only if } \llbracket p \rrbracket^{c,\pi,g}(i')(i'_t) \text{ is defined for all } i' \in \mathbf{ACC}(i)(t), \text{ otherwise undefined}
 \end{aligned}$$

To illustrate how this works, we derive a speech report under future in (81) as well as a relative clause under future in (82). In both examples, $\mathbf{OP}_\pi$ applies to the embedded proposition, shifting the perspective to the time component of an abstract index. In (81), the index is bound by the the intensional predicate *say*, whose time argument is supplied by *will*. Therefore, the perspective of the shifted present and $\ulcorner then \urcorner$ are indirectly bound by *will*. In (82), on the other hand, the main verb *meet* is extensional and thus cannot bind the index; the shifted perspective is thus directly bound by *will*. In both cases, $\ulcorner then \urcorner$ and the present are anchored to i'_t , introduced by **WOLL**. Like before, we have a conflict between their presuppositions and the assertion, resulting to semantic anomaly.

²²The proposed semantics for **WOLL** should be read as schematic, as the semantics of the future is actively debated. That is, although we specified **WOLL** as having a universal modal force (following Ogihara and Sharvit 2012), we acknowledge that various alternative proposals exist (see e.g. Cariani (2021)). We also remain neutral as to whether the future reading is part of the semantics of **WOLL** or is derived thanks to a prospective aspect.

with the present under future in English, an SoT language. We also find that, surprisingly, this LF with a deleted present is unavailable in Modern Greek, since ‘*then*’ in Modern Greek is incompatible with present-under-future. We present several possible explanations of this puzzle, but ultimately leave it open for future research. This finding adds to the cross-linguistic variation among SoT languages.

(83) Sequence-of-tense (SoT)

The tense feature of a tense c-commanded by an agreeing tense may be deleted at the Logical Form.

(84) $\llbracket \text{WOLL-PRES}_0 \lambda_1 [\text{Nate say-}t_1 [\text{OP}_\pi \text{ Erica be-} \text{PRES}_1 \text{ angry then}_3]] \rrbracket^{c,\pi,g}$

5.4 Interim summary

In this Section, we showed what goes wrong compositionally when ‘*then*’ and the present appear in the same clause. Irrespective of the language or the embedding environment where the puzzle occurs in, its causes are the same. We saw that ‘*then*’ cannot restrict the reference of the present when they share the same temporal perspective. This follows directly from our parameter-based approach, where the perspective is a parameter and thus has to be shared by ‘*then*’ and the present when they appear in the same minimal clausal domain.

Tense shift is treated uniformly as the result of a propositional operator OP_π shifting the perspective by overwriting it with an abstract evaluation index. Importantly, the index has to be bound by a higher intensional operator, which predicts that tense shift occurs only in intensional environments. Overall, the present analysis is empirically aligned with the observed asymmetry between relative clauses and speech/attitude reports (with the exception of Japanese to be addressed in Appendix A) and between the embeddings under future and past, while improving on previous analyses by offering a principled solution to the ‘*then*’-present puzzle.²³

6 Deleted tenses are not perspective-sensitive

A crucial point of this paper is the asymmetry between *shifted* present and *deleted* tense. Recall that in Modern Greek, a language with both a shifted present and a deleted past (Sharvit, 2018; Tsilia, 2021, 2022a,b), ‘*then*’ is only compatible with the latter. This has theoretical importance, since the shifted present and the deleted past were up to now assumed to be semantically equivalent (Ogihara and Sharvit, 2012). The puzzle arising in Modern Greek only with the shifted present, but not with the deleted past, teaches us that they cannot be equivalent. In this Section, we propose to model their difference in terms of sensitivity to the perspective parameter: the shifted present is sensitive to the perspective, but the deleted past is not.

The key diagnostic for deleted tense, (48), is repeated below as (85). Recall that this example shows that the most deeply embedded past has to be deleted in English, since it is not past relative to any other moment in the sentence. Notice, also, that it is compatible with *then*:

²³An alternative analysis would be to take the behavior of Japanese as a baseline, and assume all tenses (not just Japanese ones) are intensional. However, we would then over-generate readings for Modern Greek, Modern Hebrew, English and Russian. This is why we opted for this analysis which ties intensionality to tense shift.

- (85) A week ago, John said that in ten days he would say to his girlfriend that they **were** meeting **then** for the last time.

Our system provides a straightforward explanation for the contrast. Unlike the shifted present, we argue that deleted tenses are plain temporal pronouns without any perspectival presupposition:

$$(86) \llbracket \mathbf{PRES}_n \rrbracket^{c,\pi,g} = g(n)$$

$$(87) \llbracket \mathbf{PAST}_n \rrbracket^{c,\pi,g} = g(n)$$

Let us illustrate this with the following working example (which has a structure similar to (85)). Here we consider two possible LFs, as shown in (88a) and (88b), for a minimal comparison. In both LFs, the most deeply embedded past tense (**PAST**₁ and **PAST**₂, respectively) are deleted. Note that *would* is assumed to be the past tense of *will*, and thus decomposed into **WOLL** and a past tense **PAST**₁.²⁴

- (88) John think₁ a week ago that in ten days Mary would₁ say that she **missed**₂ Nate **then**₃.

- a. John think-**PAST**₁ a week ago

$$[\mathbf{WOLL-PAST}_1 \lambda_2 [\text{Mary say-}t_2 \text{ in ten days} \\ [\text{Mary miss-}\mathbf{PAST}_2 \text{ Nate then}_3]]]$$

- b. John think-**PAST**₁ a week ago

$$[\mathbf{WOLL-PAST}_1 \lambda_2 [\text{Mary say-}t_2 \text{ in ten days} \\ \mathbf{OP}_\pi [\text{Mary miss-}\mathbf{PAST}_2 \text{ Nate then}_3]]]$$

The denotation of the most deeply embedded clause without **OP**_π in (88a) is given in (89a) below. Deleted tense does not carry any presupposition; only ‘*then*’ carries its usual presupposition that $g(3)$ is disjoint from the temporal perspective. And since (88a) contains no occurrence of **OP**_π, it simply means $g(3)$ is disjoint from the time of utterance. On the other hand, if **OP**_π applies as in (88b), the interpretation of *then*₃ is shifted to be anchored to i'_t , which will bound by the higher intensional operator *say*. But importantly, here **OP**_π does not affect the interpretation of **PAST**₂ at all. As such, in both cases there is no conflict between deleted tense and ‘*then*’.

- (89) a. $\llbracket \text{Mary miss-}\mathbf{PAST}_2 \text{ Nate then}_3 \rrbracket^{c,\pi,g}$
 $= \lambda i. \mathbf{MISS}(i)(g(2))(\mathbf{M})(\mathbf{N}) \wedge \boxed{g(2) \subseteq g(3)}, \text{ only if } \boxed{\neg(g(3) \circ \pi)}$
- b. $\llbracket \mathbf{OP}_\pi [\text{Mary miss-}\mathbf{PAST}_2 \text{ Nate then}_3] \rrbracket^{c,\pi,g}$
 $= \lambda i'. \llbracket \text{Mary miss-}\mathbf{PAST}_2 \text{ Nate then}_3 \rrbracket^{c,i',g}$
 $= \lambda i'. \mathbf{MISS}(i')(g(2))(\mathbf{M})(\mathbf{N}) \wedge \boxed{g(2) \subseteq g(3)} \text{ only if } \boxed{\neg(g(3) \circ i'_t)}$

²⁴For the past tense on *would*, we could have assumed that is interpreted *de re*, relative to the time of the utterance, as opposed to deleted. But the choice will not matter here. Importantly, as discussed in Section 3, this option is not available for the past tense on *miss*, given the intended reading.

Indeed, if we compare (89) to the semantic anomaly triggered by ‘*then*’ and the present (as in (77), (81) and (82)), there is one less presupposition: the presupposition of the shifted present is missing. The only presupposition in both cases is the one carried by *then*₃, which does not contradict the asserted temporal restriction $g(2) \subseteq g(3)$. Therefore the semantic anomaly is prevented.

That being said, it should be noted that the LF in (89a) is the one that generates the reported simultaneous interpretation. For in (89b), the operator $\mathbf{OP}_\pi$ shifts the local perspective to the time of Mary’s future speech, and so the presupposition of *then*₃ forces it to denote a time disjoint from it. In that case *then*₃ can no longer be used to coherently restrict the reference of the deleted past, as intended by the simultaneous reading. In (89a), on the other hand, *then*₃ is not shifted and thus presupposed to be disjoint from the time of *utterance*, which makes it possible for it to contain the time of Mary’s speech three days from now.

As mentioned in the last Section, the deleted past is not the only possible deleted tense. The SoT rule (see (83) above) applies to any tense morpheme c-commanded by an agreeing tense morpheme, and as such it also applies to the present under **WILL** (decomposed into **WOLL** and **PRES**). As a reviewer points out, the possibility of a deleted present under future explains the compatibility of *then* with present under future in English in (30d) and (31d), repeated below as (90) and (91) respectively:

(90) In 2030, John will say that he is in jail (*%then*). (Tsilia 2021: (85))

(91) In 2030, John will be married to someone who is in jail (*then*).

(Tsilia 2021: (87), adapted from von Stechow and Heim (1997-2021): §7, (4))

The examples above escape the ‘*then*’-present puzzle, since the embedded present is deleted.²⁵ The deleted present has the entry in (86), and the LFs of the English examples are below:

(92) $\llbracket \mathbf{WOLL-PRES}_0 \lambda_1 [\text{John say-}t_1 [\mathbf{OP}_\pi \text{ he be-} \mathbf{PRES}_2 \text{ in-jail then}_3]] \rrbracket^{c,\pi,g}$

(93) $\llbracket \mathbf{WOLL-PRES}_0 \lambda_1 [\text{some (PERSON)} (\lambda_2 [\mathbf{OP}_\pi [x_2 \text{ be-} \mathbf{PRES}_3 \text{ in-jail then}_5]])] \lambda_6 [\text{John married-}t_1 x_6]] \rrbracket^{c,\pi,g}$

But if English has these LFs which escape the ‘*then*’-present puzzle under future, then Modern Greek, another SoT language, should also have a deleted present under ‘*will*’. Yet, in subsection 5.3 above, we presented the incompatibility of ‘*then*’ with the present under future in Modern Greek as an instance of the puzzle. Crucial

²⁵As the same anonymous reviewer points out, there is still a puzzle with present-under-*would* readings (Anand and Hacquard, 2008; Wurmbrand, 2014; Sharvit, 2014) being compatible with *then*, such as the following (tested with 3 speakers):

- (1) Context: A week ago I saw John and he was very concerned about his relationship. He said he would meet his girlfriend in 10 days to break up. He specifically said: “In 10 days I’ll tell her that this is the last time we’re meeting”.
A week ago, John said that in ten days he would say to his girlfriend that they are meeting then for the last time.

This is puzzling since *would* is composed of **WOLL** and **PAST**, and thus the most embedded present does not get deleted. We leave a more thorough investigation of such examples for future research, but we point out that the present here could actually be interpreted as a planned future. In other words, the speaker could be taking John to have said “I am breaking up with my girlfriend then/in 10 days”, since John has a plan to do so. In this utterance the present is only morphological, since semantically it is interpreted as a future or as the future-oriented *plan to*. As Tsilia (2021) has shown, ‘*then*’ can be compatible with present morphology as long as this morphology is not semantically present (e.g., historical present). We also see this with present-under-future in English, where *then* is compatible with a deleted present, which is morphologically but not semantically a present.

to this argument was the idea that the present is interpreted, and is thus perspective-sensitive, conflicting with the perspectival presupposition of 'then'. If a deleted present is allowed in English, what blocks it in Modern Greek? In other words, why does Modern Greek pattern with Modern Hebrew and Russian rather than English when it comes to present-under-future?

This subtle difference between two SoT languages, such as English and Modern Greek has not been noted before, and it indicates that there is more cross-linguistic variation than we initially thought within SoT languages. We already noted that an SoT language can also have a shiftable present, like Modern Greek, and now we observe that having a shiftable present interacts with the language's ability to delete the present in a present-under-future configuration. More specifically, we find the following generalization:²⁶

(94) New puzzle about deleted present:

If an SoT language also has a shiftable present, the present cannot be deleted.

We entertain two hypotheses to explain this. First of all, there could be two versions of the SoT rule, one that applies to all agreeing morphemes (see (83) repeated below as (95)), and one that applies only to past tense morphemes (see (96)). English would have (95), while Modern Greek would have (96).²⁷ This explanation, however, leaves open the possibility that a language has (95) and a shiftable present, therefore disproving the generalization in (94). It could be an option if we find a language that has SoT and a shiftable present, like Modern Greek, but where 'then' is compatible with present-under-future, unlike Modern Greek.

(95) Sequence-of-tense (SoT)

The tense feature of a tense c-commanded by an agreeing tense may be deleted at the Logical Form.

(96) Past Tense Sequence-of-tense (SoT)

The tense feature of a past tense morpheme c-commanded by an agreeing past tense morpheme may be deleted at the Logical Form.

Another possibility, which would better explain the generalization in (94), is that there is a principle forcing the present in Modern Greek (but not in English) to always be interpreted. This principle would be as follows:

(97) Interpret the present!

In a language with a shiftable present, always interpret the present, do not ever delete it.

This would predict that the Modern Greek present has to be interpreted (either by being shifted or interpreted indexically/*de re*). In other words, the Modern Greek present cannot be deleted despite the rule in (95). In that case, this principle has to over-ride the SoT rule in (95). It would also account for the generalization in

²⁶Our sample of an SoT language with a shiftable present is limited to Modern Greek. More research is thus needed to verify this generalization, testing whether 'then' is compatible with present-under-future in languages that behave otherwise identically to Modern Greek.

²⁷It would not be enough to say that we only have the SoT rule in (96), since in English we know from the compatibility of 'then' with the present that the latter can be deleted (and therefore we need (95) at least for English).

(94), since we would predict that in any SoT language with a shiftable present ‘*then*’ will be incompatible with present-under-future. To choose between these two hypotheses, we would need to find more languages like Modern Greek, with both a deleted past and a shiftable present. We leave this open for future research.

Having shown that ‘*then*’ is compatible with deleted tense, and having cashed that out in our system in terms of deleted tenses being perspective insensitive, we will now present evidence that the perspective parameter is not redundant, and must be distinct from both the context and the index.

7 Perspective is not Context or Index

So far, we have accounted for the data discussed in Sections 2 and 3 by arguing that, contrary to what has been assumed up to now, deleted and shifted tense do not have the same interpretation. Instead, we proposed that although shifted present is sensitive to the perspective parameter π , deleted tenses are not. But an important question has been left unaddressed—what exactly is the perspective parameter?

In this Section, we will provide a partial answer by pointing out what the perspective is *not*. In the interest of theoretical simplicity, one may be tempted to identify the perspective with (the temporal component of) other interpretation parameters, such as the context or the index. We argue that this cannot be the case, and thus the perspective has to be a *distinct* parameter, especially designated for the interpretation of tenses.

7.1 Perspective is not Context

Suppose that π is in fact just the time component c_t of the context parameter. Then, tenses and ‘*then*’ are essentially temporal indexicals like ‘*now*’, ‘*tomorrow*’ and ‘*yesterday*’ whose interpretations are dependent on the utterance context, as illustrated below:

- (98) a. $\llbracket \text{PRES}_n \rrbracket^{c,g} = g(n)$ only if $\boxed{g(n) \circ c_t}$, otherwise undefined.
 b. $\llbracket \text{then}_n \rrbracket^{c,g} = g(n)$ only if $\boxed{\neg(g(n) \circ c_t)}$, otherwise undefined.
 c. $\llbracket \text{now} \rrbracket^{c,g} = c_t$

In this case, assuming that tense shift is due to the shift of c_t makes the additional prediction that languages that allow tense shift would allow for concomitant indexical shift as well. This is not the case, see e.g. the example below from Modern Greek:

- (99) To 1960, o Yanis iksero oti i Maria **ine** omorfi (#**tora**).
 DET 1960, DET Yanis know.PAST that DET Maria **be.PRES** beautiful **now**
 ‘In 1960, Yanis knew that Maria was (LIT.: is) beautiful.’ Modern Greek (Tsilia 2021: (100))

Without the temporal indexical *tora* ‘now’, the embedded present in (99) can receive a shifted reading where it reports Yanis’ past knowledge in 1960 that Maria was beautiful at that time. However, adding the temporal indexical *tora* ‘now’ forces a rather marginal reading that Yanis clairvoyantly predicted Maria’s

state of beauty at the time of utterance. In other words, the temporal indexical *tora* ‘now’ does not shift, has to refer to the time of utterance, and therefore blocks the shifted interpretation of the embedded present. The same holds for Modern Hebrew and Japanese, as shown in (100) and (101):

- (100) Lifney alpayim šana, Yosef **xašav** še (#**axšav**) Miriam **ohevet** oto.
 before two-thousand year Yosef **think.PAST** that **now** Miriam **love.PRES** him
 ‘Two thousand years ago, Yosef thought that Miriam loved (LIT.: loves) him.’ Modern Hebrew
- (101) ninenn mae, Yusuke-wa Yukiko-ga (#**ima**) ninnshinn **shite-iru** to **shitteita**
 2-years before Yusuke-TOP Yukiko-NOM **now** pregnant **be-PRES** with **know.PST**.
 ‘2 years ago, Yusuke knew that Yukiko was (LIT.: is) pregnant.’ Japanese

Relatedly, while indexical shift is generally recognized as a phenomenon restricted to speech/attitude reports (e.g. both Anand (2006) and Deal (2020) propose a shifting operator *selected* by certain speech/attitude predicates), tense shift, as we have seen, may occur in non-attitudinal environments as well, in languages like Japanese. Therefore even if the underlying mechanisms are similar, tense shift and indexical shift should not follow from the exact same operation. We take these to be strong evidence for distinguishing π and c_t .

That being said, we would like to point out that separating π and c_t does not mean that they are completely independent through the course of semantic evaluations, and their respective values may be correlated in a systematic way. We will return to this point shortly.

7.2 Perspective is not Index

Now suppose that the perspective is just the temporal component of the evaluation index, i.e. $\pi = i_t$. This is roughly the position taken by many previous approaches (e.g. Ogihara 1996; Abusch 1997; Kratzer 1998; Kusumoto 1999; Ogihara and Sharvit 2012 a.o.). In our system this would mean that the semantics of tenses and ‘*then*’ would denote partial functions from the index to a time interval:

- (102) a. $\llbracket \text{PRES}_n \rrbracket^{c,g} = \lambda i : g(n) \circ i_t . g(n)$
 b. $\llbracket \text{PAST}_n \rrbracket^{c,g} = \lambda i : g(n) < i_t . g(n)$
 c. $\llbracket \text{then}_n \rrbracket^{c,g} = \lambda i : \neg(g(n) \circ i_t) . g(n)$

The first problem of this is that identifying π with i_t has the same effect as making the shifting operator OP_π *obligatory*. This immediately predicts that tense shift is obligatory in speech/attitude reports, as shown in (103). Since the embedded tense is now anchored to the index and the speech/attitude verb always binds the evaluation index of its complement, a shifted reading becomes unavoidable.

- (103) Nate said Erica is angry. (Lit. from Modern Greek)
 LF: $\llbracket \text{Nate say-PAST}_k [\text{Erica be-PRES}_n \text{angry}] \rrbracket^{c,g}$
 a. $\llbracket \text{Erica be-PRES}_n \text{angry} \rrbracket^{c,g} = \lambda i' . \text{ANGRY}(g(n))(E)(i'_w)$ only if $g(n) \circ i'_t$

- b. $LF \rightsquigarrow \lambda i. \forall i' \in \mathbf{SAY}(g(k))(\mathbf{N})(i_w). \mathbf{ANGRY}(g(n))(\mathbf{E})(i'_w)$
only if $\forall i' \in \mathbf{SAY}(t)(x)(i_w). g(n) \circ i'_t (= g(k))$

This is an undesirable result. Even in languages where tenses *can* shift, the shifting is not obligatory. For example, certain temporal adverbials may force the present to be interpreted with respect to the time of utterance. This is shown in (104)—even though Modern Greek allows for a shifted present in reported speech under future, the embedded present can still be interpreted relative to the time of utterance if forced by the non-shiftable indexical *tora* ‘now’ (cf. (99)).²⁸

- (104) To 2030, i Maria **tha** pi oti **ine** sti filaki tora.
DET 2030, DET Maria **FUT** say.PFV that **be.PRES** to-the jail now
‘In 2030, Maria will say that she is in jail now.’ Modern Greek

Another reason not to identify the perspective with the temporal component of the index is that it would make the wrong prediction for deleted tense. Recall the possible LF of deleted past with ‘*then*’ in (89):

- (89) a. $\llbracket \text{Mary miss-}\mathbf{PAST}_2 \text{ Nate then}_3 \rrbracket^{c,\pi,g}$
 $= \lambda i. \mathbf{MISS}(i)(g(2))(\mathbf{M})(\mathbf{N}) \wedge \boxed{g(2) \subseteq g(3)}, \text{ only if } \boxed{\neg(g(3) \circ \pi)}$
b. $\llbracket \mathbf{OP}_\pi [\text{Mary miss-}\mathbf{PAST}_2 \text{ Nate then}_3] \rrbracket^{c,\pi,g}$
 $= \lambda i'. \llbracket \text{Mary miss-}\mathbf{PAST}_2 \text{ Nate then}_3 \rrbracket^{c,i',g}$
 $= \lambda i'. \mathbf{MISS}(i')(g(2))(\mathbf{M})(\mathbf{N}) \wedge \boxed{g(2) \subseteq g(3)} \text{ only if } \boxed{\neg(g(3) \circ i'_t)}$

Recall also that the simultaneous reading is obtained via the LF in (89a), where $\mathbf{OP}_\pi$ is absent. This is because when $\mathbf{OP}_\pi$ is present, the presupposition of *then*₃ is that it denotes a time distinct from the *shifted* perspective. Thus, when $\mathbf{OP}_\pi$ is present, *then*₃ cannot refer to a time simultaneous with the embedded verb. Instead, we need the LF in (89a) without the perspective shifting operator in order to get the simultaneous reading. But by identifying π with i_t and predicting obligatory tense shift, we lose the simultaneous reading. Indeed, if the perspective simply *is* the temporal component of the index, then both LFs would generate the same shifted reading as in (105), where *then*₃ has no way of denoting a time simultaneous with the time of the missing.

- (105) a. $\llbracket \text{Mary miss-}\mathbf{PAST}_2 \text{ Nate then}_3 \rrbracket^{c,g}$
 $= \lambda i. \mathbf{MISS}(i)(g(2))(\mathbf{M})(\mathbf{N}) \wedge \boxed{g(2) \subseteq g(3)}, \text{ only if } \boxed{\neg(g(3) \circ i_t)}$
b. $\llbracket \mathbf{OP}_\pi [\text{Mary miss-}\mathbf{PAST}_2 \text{ Nate then}_3] \rrbracket^{c,\pi,g}$
 $= \lambda i'. \llbracket \text{Mary miss-}\mathbf{PAST}_2 \text{ Nate then}_3 \rrbracket^{c,g}$
 $= \lambda i'. \mathbf{MISS}(i')(g(2))(\mathbf{M})(\mathbf{N}) \wedge \boxed{g(2) \subseteq g(3)}, \text{ only if } \boxed{\neg(g(3) \circ i'_t)}$

²⁸ An anonymous reviewer points out that the indexical reading of the present could be achieved in a perspective-as-index approach if we assume a distinguished variable t^* at the left periphery (von Stechow, 1995; Kusumoto, 2005) or a now operator (Kamp, 1967). Although this is a viable alternative that should be explored in future research, we prefer our system for reasons of theoretical simplicity. The t^*/now operator would have to be constrained in some way, or else, if always there, it would often be redundant. Also, we do not think that adding a t^*/now is necessarily simpler than adding a perspective, especially since the latter can explain a range of phenomena as will be seen in the next subsection (see also Sudo (2015) for locative perspectives for example).

In sum, reducing the perspective to either the temporal component of the context or the index will lead to undesirable predictions, and it is an important empirical claim of the paper that the interpretation of tenses calls for a *distinct* perspective parameter π .

7.3 More on the notion of perspective

Having established that we need a perspective parameter π , we may now ask whether π is always independent. It could be that π does not interact with other parameters at all, or that in certain contexts there are implicational hierarchies. Indeed, we already saw that in matrix clauses the perspective is co-valued with the temporal component of the context. In other words, by default, $\pi = c_t$. So, the perspective and the context parameters do interact. Are there other contexts where the perspective is forced to be co-valued with the temporal component of the context? We suggest that there are.

Kusily (2017) explored the interaction between temporal indexicals, such as *today*, *yesterday* and *tomorrow*, and the shifted present. He observed that in Russian the shifted present can only co-occur with the likewise shiftable temporal adverbial *segodnja* ‘today’, but not with the indexical *včera* ‘yesterday’, despite them both referring to the day before the time of utterance. This is illustrated below:

- (106) *Context: Today is Tuesday. On Monday (i.e. one day earlier), Vanja utters: “Masha is sick today”.*
- a. Vanja skazal, čto Maša **segodnja bol’na**.
 Vanja said that Masha **today be.PRES-sick**
 “Vanja said that Masha was sick on that day.”
 - b. #Vanja skazal, čto Maša **včera bol’na**.
 Vanja said that Masha **yesterday be.PRES-sick**
 Intended: “Vanja said that Masha was sick yesterday.” Russian (Kusily 2017: (20))

Similar effects are observed under future, an environment that generally allows tense shift. In the English and Japanese examples below the embedded past is shifted and is only past relative to the ‘now’ of the attitude holder while being in the future with respect to the time of utterance. In such an environment where the tense is shifted, it is only compatible with a shiftable adverbial like *two days before*, but not with an indexical one like *tomorrow*.

- (107) *Context: Today is Monday. Mary arrives on Tuesday. John talks about Mary’s arrival on Thursday.*
- a. John will say that Mary **arrived two days before**.
 - b. #John will say that Mary **arrived tomorrow**. (Kusily 2017: (1))
- (108) *Context: Today is Monday. Mary arrives on Tuesday. John talks about Mary’s arrival on Thursday.*
- a. Kazuko-wa [**hutu-ka-mae-ni** Tokyo-ni it-ta] otoko-ni a-w(-daroo).
 Kazuko-TOP **two-day-before-in** Tokyo-to **go-PAST** man-DAT meet-PRES(-will)
 ‘Kazuko will meet a man who went to Tokyo two days before.’

- b. #Kazuko [asu Tokyo-ni it-ta] otoko-ni a-w(-daroo).
 Kazuko-TOP tomorrow Tokyo-to go-PAST man-DAT meet-PRES(-will)
 Japanese (Kusily 2017: (2))

These data pattern well with the SHIFT TOGETHER effect, as we have seen with ‘*then*’ and the present. When the indexical *včera* ‘yesterday’ is used in Russian, *asu* ‘tomorrow’ in Japanese, or *tomorrow* in English, the embedded tense is also interpreted indexically rendering the example infelicitous in the context. On the other hand, when the shiftable *segodnja* ‘today’ is used in Russian, *hutu-ka-mae-ni* ‘two days before’ in Japanese, or *two days before* in English, the embedded tense is also shifted. If the parameters are entirely independent in these environments, then we do not predict this SHIFT TOGETHER effect, since the tenses are evaluated with respect to the perspective and the temporal adverbials with respect to the context or the index parameters. Mismatches between the parameters are allowed, so we overgenerate readings for (106b), (107b) and (108b). To see this, take the example of (106b), repeated below as (109):

- (109) #Vanja skazal, čto Maša včera bol’na.
 Vanja said that Masha yesterday be.PRES-sick
 Intended: “Vanja said that Masha was sick yesterday.” Russian (Kusily 2017: (20))
- a. $\llbracket \text{yesterday} \rrbracket^{c,\pi,g} = \text{the day before } c_t$
 b. $\llbracket \text{Vanja say-PAST}_1 [\text{OP}_\pi \text{ Masha be-PRES}_2 \text{ sick yesterday}] \rrbracket^{c,\pi,g}$

We currently allow for the LF in (109b), which contains the perspective-shifting operator OP_π but not the context-shifting operator. Given the standard semantics in (109a) for *yesterday*, where it picks out the day before c_t , the embedded present PRES_2 can be shifted, while the temporal indexical is interpreted with respect to the context. To block this LF, we need to establish a dependency between perspective and context, along the lines of the following constraint:

- (110) When an overt temporal indexical occurs within a minimal clausal domain, the temporal perspective π is co-valued with the temporal component of the context c_t .

But in fact things are even more complicated. Not all temporal indexicals block a co-clausal shifted tense. For instance, Kusily (2017) reported that (111), with the non-shifted indexical *today*, is felicitous in the given context. Since Charlie’s arrival is expected *later* today, the embedded past can only be interpreted relative to a shifted perspective that presumably coincides with Alice’s future speech. The fact that it is compatible with *today* clearly violates (110).

- (111) Context: Charlie is expected to arrive today, although he hasn’t at the time of utterance.
 Alice will tell Bob tomorrow that Charlie arrived today.

Thus, (110) will need to be further refined, or else *today* could be shown to marginally have shiftable

usages.²⁹ We leave a full exploration of the exact dependencies between parameters for later, but we hope to have opened an avenue on the interaction between (non-)shiftable adverbials and tense. Another interesting direction for future research is the dependency between the perspective and the index, as well as non-temporal usages of the perspective. For other perspective-sensitive expressions outside the temporal domain (Sudo, 2015), a more thorough treatment of the perspective parameter is needed, containing a speaker, an addressee, and a world, parallel to the context and the index.

On the other hand, temporal adverbials other than ‘*then*’ may be analyzed as being perspective-sensitive. One example is the shiftable *now* in Russian. Vostrikova (2019) discusses interesting data where *sejčas* ‘now’ in Russian is incompatible with an embedded past interpreted *de re*, targeting a simultaneous reading:

- (112) Kogda ja govorila s nej tri goda nazad, Tanja skazala, čto ona byla (*sejčas)
 When I talk.PAST with her three years ago Tanja say.PAST that she be.PAST now
 beremenna.
 pregnant
 Intended: ‘When I talked to her three years ago, Tanja told me that she was pregnant (at that time).’
 Russian (Vostrikova 2019: (6))

Unlike English *now*, *sejčas* ‘now’ in Russian can be shifted, referring to a time other than the time of utterance. This is shown in (113) below, where *sejčas* is compatible with a shifted present.³⁰

- (113) Kogda ja govorila s nej tri goda nazad, Tanja skazala, čto ona (sejčas) ∅ beremenna.
 When I talk.PAST with her three years ago Tanja say.PAST that she now be.PRES pregnant
 ‘When I talked to her three years ago, Tanja told me that she was pregnant (at that time).’
 Russian (Vostrikova 2019: (5))

²⁹Kusily argues that temporal indexicals as well as *then* ‘force feed’ their interpretation parameters (c_t for indexicals and i_t for *then*) to the co-clausal tense. He further distinguishes between what he calls (temporally) *oriented* adverbials (such as *yesterday*, *tomorrow*, *then*, *two days ago*) and *non-oriented* ones (such as *today*, *two days before*) claiming that *oriented* adverbials are (lexically) equipped to ‘force feed’ their parameter to the co-clausal tense. However, since he defines *oriented* adverbials using their incompatibility with a co-clausal tense, and uses this notion to then explain this incompatibility, the orientedness criterion becomes somewhat circular.

³⁰There is within-speaker variation with respect to whether Russian speakers have a shiftable ‘now’. We tested (1) with two consultants, one of which had an indexical ‘now’, while the other could access a shifted reading:

- (1) V 2000m godu Ivan znal, čto Maša (%sejčas) beremenna.
 In 2000 year Ivan know.PRES that Masha now pregnant
 ‘In 2000, Ivan knew that Masha was (LIT.: is) pregnant.’

More specifically, the consultant with the indexical reading found (1) odd, and the only possible reading they could access involved clairvoyance (like in Greek, Japanese). The other consultant initially had an indexical reading of ‘now’, but after some thought could also access a shifted reading of both ‘now’ and the embedded present, as in the following (Katya Murgonova, p.c.):

- (2) Ivan znal, čto sejčas Maša beremena i nuzhno ekonomit’ den’gi
 Ivan know.PAST that now Masha pregnant and need-to save money
 ‘Ivan knew that Masha was (LIT.: is) pregnant then (LIT.: now) and that they should save up money.’

The consultant also mentioned that under the shifted reading, ‘now’ cannot be prosodically focused. We treat some Russian speakers as having the entry of the English/Greek indexical ‘now’, and some as having the shiftable *sejčas* ‘now’ Vostrikova describes.

How can we look at these data under the perspective parameter lens? Recall the entry for the past from (70), repeated below as (114). Just like $\lceil \text{then} \rceil$, *sejčas* ‘now’ can also be given a perspective-sensitive denotation:

(114) $\llbracket \text{PAST}_n \rrbracket^{c,\pi,g} = g(n)$ only if $g(n) < \pi$, otherwise undefined

(115) $\llbracket \text{sejčas}_n \rrbracket^{c,\pi,g} = g(n)$ only if $g(n) \circ \pi$, otherwise undefined

The incompatibility between the past and *sejčas* ‘now’ when they share the same perspective is then derived as a semantic anomaly, just like the $\lceil \text{then} \rceil$ -present puzzle. The simultaneous reading of the embedded past is derived with the following LF, where OP_π does not apply and the reference of the embedded past coincides with that of the matrix past (i.e. $g(2) = g(1) < \pi$):

(116) $\llbracket \text{Tanja say-PAST}_1 [\text{she be.PAST}_2 \text{ pregnant}] \rrbracket^{c,\pi,g}$

Finally, we would like to point out a difference between the perspective parameter and the context one, which should be explored in future research. Operator-based analyses of indexical shift make the nice prediction that a shiftable indexical cannot pick its reference from a context c if there is an intervening context c' from which another indexical picks its reference, as illustrated below:

(117) [Andrew:] Ali_A mi_U-ra va ke Hesen_H **to**_{U-ra} va $\epsilon z_{H,A,*U}$ braye Rojda-o
 Ali me-to said that Hesen **you**-to said **I** brother Rojda-GEN
 ‘Ali said to Andrew that Hesen said to Andrew that {Hesen, Ali, *Andrew} is Rojda’s brother.’
 Zazaki (Anand and Nevins 2004: (32))

There are two shifted indexicals in (117): the embedded second-person indexical *to* ‘you’ that is shifted to the addressee in the reported speech (who happens to be the actual speaker Andrew), and a more deeply embedded first-person indexical ϵz ‘I’. Crucially, with the intermediate indexical *to* ‘you’ *shifted*, the lowest indexical ϵz ‘I’ can no longer draw reference from the matrix utterance context; instead, it has to be interpreted relative to the intermediate speech context (whose speaker is Ali), or to be further shifted, referring to the speaker of the lowest reported speech, namely Hesen. This is because once the operator overwrites the context parameter, the information about the matrix context is lost. This is known as the NO INTERVENING BINDER constraint. As a reviewer points out, we make the same prediction in the temporal domain, since once we shift the perspective we can no longer retrieve the original perspective parameter. To test this prediction, consider the following examples containing two presents under future:³¹

(118) John will announce that he **is** working with someone who **is** working with the government.

(119) O Yanis tha anakinosi oti dhulevi me kapion pu
 DET Yanis FUT announce.PFV COMP work.PRES.IMPFV.3SG with someone.ACC who
 dhulevi stin kivernisi.
 work.PRES.IMPFV.3SG to-the government

³¹We thank Amy Rose Deal for helpful discussions on this and similar issues during WCCFL, as well as an anonymous reviewer who also points this out.

- a. ✓Yanis: “I am working w/ someone who **is** working for the government” (both shifted)
- b. ??Yanis: “I am working w/ someone who **was** working for the government” (shifted, indexical)
- c. ✓Yanis: “I **was** working w/ someone who **was** working for the government” (both indexical)
- d. ✗Yanis: “I **was** working w/ someone who **is** working for the government” (indexical, shifted)

The current operator-based analysis similarly predicts that both presents should be shifted or both should be indexical; it does not allow for mixing of perspectives. This correctly rules out (119d), while also predicting ungrammaticality for (119b). In (119b) the first present is shifted, while the second is read indexically; this reading is only available in a scenario where the second present is true at the time of the utterance *and* at the time of the speech as in (120a) and (120b) below. It is impossible if the second present is only indexical and is not true at the time of the speech, as shown in (120c) and (120d). Thus, a SHIFT TOGETHER effect is at play here.

(120) O Yanis tha anakinosi oti dhulevi me kapion pu
 DET Yanis FUT announce.PFV COMP work.PRES.IMPFV.3SG with someone.ACC who
 dhulevi stin kavernisi.
 work.PRES.IMPFV.3SG to-the government

- a. First present indexical and shifted, Second present indexical and shifted ✓
 Yanis has been working for many months with the minister of agriculture. Yanis will announce it at the company’s meeting next Fall.
- b. First present only shifted, Second present indexical and shifted ✓
 Yanis will start working with the minister of agriculture in 2 months. He will announce it at the company’s meeting next Fall.
- c. First present only shifted, Second present only indexical ✗
 The current minister of agriculture plans to resign from his government position and start working with Yanis. Yanis and the current minister have signed a contract, and will start working together in 2 months. Yanis plans to announce it at the company’s meeting next Fall.
- d. First present indexical and shifted, Second only indexical ✗
 The current minister of agriculture is working with Yanis, whose company is a government contractor. The minister plans to resign from his government position and continue working exclusively with Yanis. Yanis plans to announce it at the company’s meeting next Fall.

We have delved deeper into the perspective parameter and discussed ways in which it can be further explored, in the temporal domain and beyond. In Appendix B, we discuss another potential extension of the perspective parameter to the modal domain.

8 Conclusion

This paper delved into the observation that ‘*then*’ is incompatible with a shifted present, but compatible with deleted tense. To account for this puzzle, we proposed a new theory of tense and tense shift, centered around the notion of a temporal *perspective*. The temporal perspective is an interpretation parameter to which the interpretations of tenses are anchored; it can be shifted by a propositional perspective-shifting operator resulting in tense shift. We have shown that this new theory provides an elegant solution to the ‘*then*’-present puzzle, while also capturing the difference between shifted present and deleted tense. Although we only scratched the surface of the nature of the temporal perspective, we emphasized that it should be distinguished from more familiar parameters such as the context and the evaluation index. Formally, our system mirrors the operator-based analysis of indexical shift (Anand and Nevins 2004; Anand 2006), predicting a SHIFT TOGETHER effect for tenses. We hope that future research will incorporate tense to the growing body of literature establishing implicational hierarchies of shiftable, context- and perspective-sensitive expressions cross-linguistically (e.g., Deal 2017, 2020, Sundaresan 2018, 2021).

Statements and Declarations

The authors have no competing interests to declare that are relevant to the content of this article.

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Appendix A: Shifted present in Japanese relative clauses

Japanese is the only language examined in this paper that allows the present to shift in an extensional environment, namely in relative clauses under past. Usually, tense shift occurs in intensional environments, either under a speech/attitude verb or under future. Japanese is the only exception. How exactly the Japanese present can shift in extensional environments is an independent puzzle from the ‘*then*’-present puzzle, which we do not attempt to solve here. However, the ‘*then*’-present puzzle could be a useful clue towards its solution. More specifically, the ‘*then*’-present puzzle persists in Japanese relative clauses. This in turn suggests, that whatever is responsible for the puzzle in intensional environments in all these other languages (including in Japanese attitude reports) is also responsible for the puzzle here. In other words, the persistence of the puzzle suggests that the same tense shifting mechanism is at work in Japanese relative clauses. We will therefore propose a similar explanation for the ‘*then*’-present puzzle in Japanese relative clauses, provided that the present has an (independent from ‘*then*’) way to shift in these environments.

Let us start with the following pseudo-glossing as a working example:

- (121) Nate met someone who is in jail (*then). (Lit. from Japanese)
 Intended: Nate met someone who was in jail (at the time they met).

Like before, the embedded present shifts thanks to $\mathbf{OP}_\pi$, which composes with the relative clause. The difficulty here is that the main verb *meet* is extensional, and therefore cannot bind the index. How can the higher tense bind the local perspective at the absence of an attitude verb? For the purposes of this paper,

we propose that tenses in Japanese are themselves intensional and can thus bind the temporal index.³² More specifically, we assume that Japanese tenses (i) specify the time of the event/state, as tenses usually do, and (ii) bind the time component i_t of the index i used to evaluate their prejacent. It is this latter function that separates Japanese tenses from other tenses we have seen so far. In (122) and (123) below, we provide a lexical entry for the Japanese present and past. They are characterized as partial functions that take as arguments a time interval t (which has to satisfy a perspectival presupposition), an intensional temporal property P (their untensed prejacent), and an evaluation index i , and return true iff P holds for t when evaluated at the index $i[i_t \mapsto t]$, which is identical to i except that the time component i_t is updated to t .³³

$$(122) \quad \llbracket \mathbf{PRES}_{\text{JAP}} \rrbracket^{c,\pi,g} : \langle i, \langle \langle \kappa, \langle i, t \rangle \rangle, \langle \kappa, t \rangle \rangle \rangle := \lambda t_i : t \circ \pi. \lambda P_{\langle \kappa, \langle i, t \rangle \rangle} . \lambda i_\kappa . P(i[i_t \mapsto t])(t)$$

where $i[i_t \mapsto t]$ an evaluation index just like i , except that i_t is substituted with t (same below).

$$(123) \quad \llbracket \mathbf{PAST}_{\text{JAP}} \rrbracket^{c,\pi,g} : \langle i, \langle \langle \kappa, \langle i, t \rangle \rangle, \langle \kappa, t \rangle \rangle \rangle := \lambda t_i : t < \pi. \lambda P_{\langle \kappa, \langle i, t \rangle \rangle} . \lambda i_\kappa . P(i[i_t \mapsto t])(t)$$

The LF of the working example (121) is given in (124). It makes explicit that (i) the matrix past tense takes scope over the untensed component of the root clause (similarly, the embedded present tense takes scope over the untensed component of the relative clause), (ii) the generalized quantifier in the object position containing the relative clause is QR'd, but still under the scope of the matrix past, and (iii) $\mathbf{OP}_\pi$ is applied at the propositional node of the relative clause.

$$(124) \quad * \text{Nate met someone who is in jail then.}$$

$$\begin{aligned} & \llbracket \mathbf{PAST}_{\text{JAP}} - t_0 \rrbracket \\ & \lambda_1 [\text{some (person)} \\ & \quad \lambda_2 [\mathbf{OP}_\pi [\mathbf{PRES}_{\text{JAP}} - t_3 [\lambda_4 [x_2 \text{ be-} t_4 \text{ in jail then}_5]]]] \\ & \quad \lambda_6 [\text{Nate meet-} t_1 x_6]]]^{c,\pi,g} \end{aligned}$$

The derivation of the generalized quantifier restricted by the relative clause is in (125). At step ① $\mathbf{PRES}_{\text{JAP}}$ takes the time argument t_3 , which overlaps with the (local) perspective, per the entry in (122). Then at step ②, the denotation of t_3 , namely $g(3)$, is fed to the predicate *be in jail*, while also used to substitute the temporal component of the index i_t . Thus, the relative clause ends up being evaluated at $g(3)$. At step ③, $\mathbf{OP}_\pi$ applies and overwrites the perspective parameter with an abstracted index i' , as usual. This leads to the shifting of ‘then’ and the present. Recall that this is an unattested LF of an ungrammatical sentence. This is why a semantic anomaly is predicted, the same one we have been seeing with all combinations of ‘then’ with the

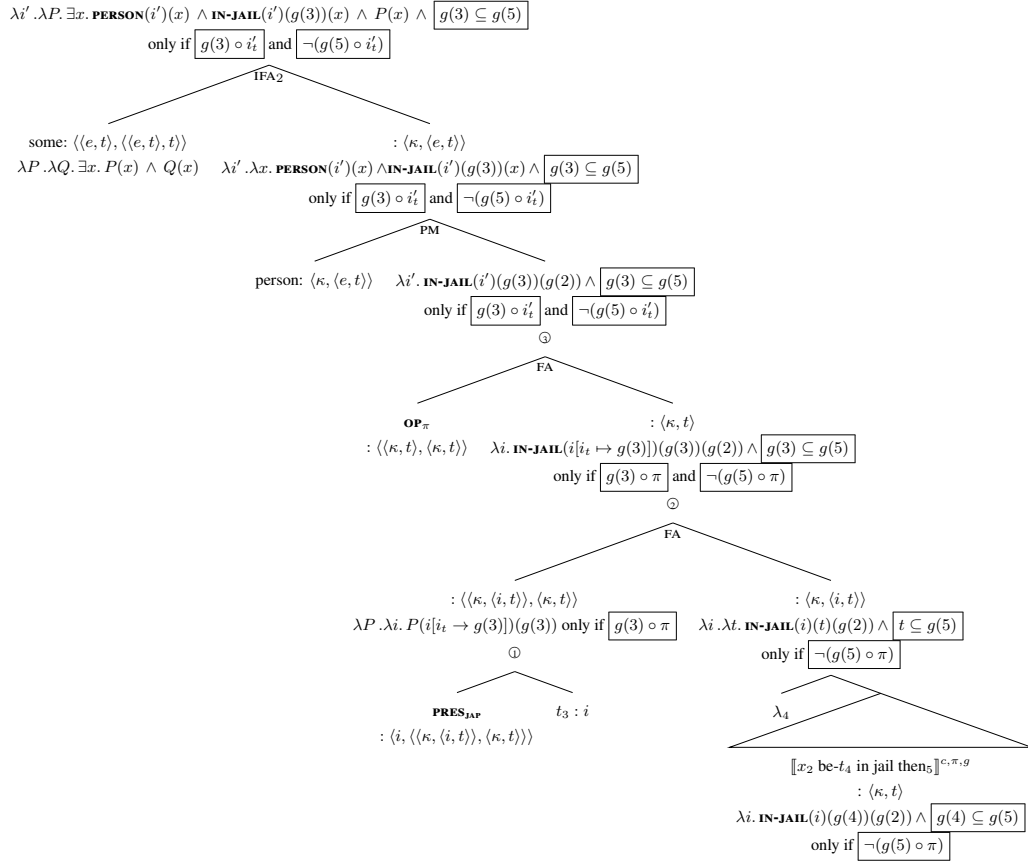
³²One does not need to agree with this solution to accept our theory of tense shift. Whatever other solution works to shift the Japanese present in an extensional environment can also be implemented in our system.

³³This analysis follows in spirit many existing theories that attempt to account for tense shift in Japanese. For instance, Ogiwara (1996, 2022) argued that Japanese tenses are purely relative (e.g. should be taken as tripartite relations of two time arguments and a temporal property), and Ogiwara and Sharvit (2012); Sharvit (2014) suggested that Japanese tenses are quantificational, as opposed to English pronominal tenses, while tense shift is achieved through the higher tense taking scope over the shifted one. The only reason we do not directly use one of these existing accounts is that they often adopt (explicitly or implicitly) a framework that treats temporal anchors as pronouns. As we argued for in Section 2.3, a parametric (as opposed to a pronominal) account of the temporal anchor is best suited to account for the ‘then’ -present puzzle.

present. More specifically, $\lceil \text{then} \rceil$ and the present share the same temporal perspective i'_t , and a contradiction arises between presupposed and asserted content.³⁴ The rest of the derivation is standard, since the quantifier has the usual type $\langle \langle e, t \rangle, \langle \langle e, t \rangle, t \rangle \rangle$, and the relative pronoun *who* is treated as a lambda-abstractor λ_2 .

(125) someone who is₃ in jail then₅

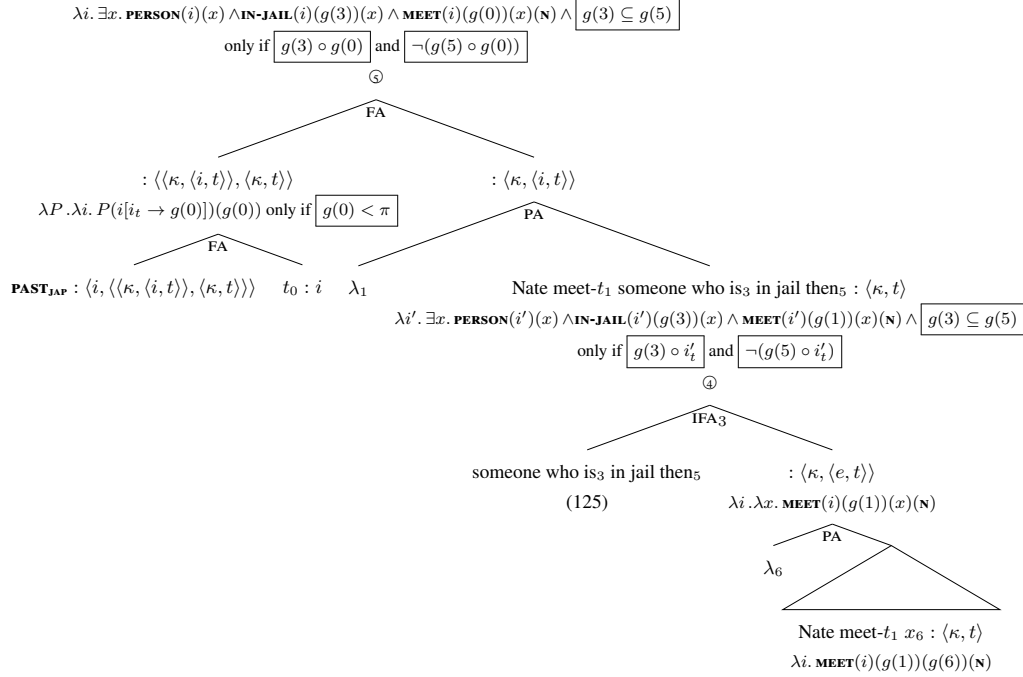
$$\llbracket \text{some (person)} \lambda_2 [\mathbf{OP}_\pi [\mathbf{PRES}_{\text{JAP}} - t_3 [\lambda_4 [x_2 \text{ be-} t_4 \text{ in jail then}_5]]]] \rrbracket^{c, \pi, g} : \langle \kappa, \langle \langle e, t \rangle, t \rangle \rangle$$



The final steps of the derivation are in (126). The quantifier takes scope over the untensed matrix clause (step ④), and the result combines with the matrix past in step ⑤. The matrix past tense is evaluated with respect to the time of utterance $g(0)$, situating the meeting event prior to $g(0)$. Finally, because we assume that Japanese tenses can bind indices, the past binds the time component of the index, i'_t , replacing it with $g(0)$. Thus, $\lceil \text{then} \rceil$ and the embedded present end up being both anchored to $g(0)$. Since they have the same anchor, and conflicting presuppositions, the same semantic anomaly as before arises.

³⁴Note that since $\mathbf{OP}_\pi$ re-abstracts over the evaluation index that the embedded present had already abstracted over, the effect of the first abstraction is essentially lost.

(126) $\llbracket \text{Nate met}_0 \text{ someone who is}_3 \text{ in jail then}_5 \rrbracket^{c, \pi, g} : \langle \kappa, t \rangle$



Making Japanese tenses able of binding indices is a technical solution, compatible with our system, to shift the present in an otherwise extensional environment. However, as already mentioned, we are not wed to this solution. The point that we want to make in this paper is that $\lceil \text{then} \rceil$ and the present have conflicting presuppositions, while $\lceil \text{then} \rceil$ and deleted past do not. And for the Japanese present, what this amounts to saying is that a shifted present will always be compatible with $\lceil \text{then} \rceil$ no matter how it shifted.

Appendix B: Further Extensions

We show that the $\lceil \text{then} \rceil$ -present puzzle extends straightforwardly to a present tense embedded under a modal. Then, we discuss a case of a morphological present in Modern Greek, which is semantically non-past, and thus is predicted to be compatible with $\lceil \text{then} \rceil$.

Extensions to present embedded under modals

We would like to bring up an interesting extension to the puzzle in present embedded under modals. As Condoravdi (2002) shows, modals can be future-oriented. For example, the following sentence in English is ambiguous between a present- and a future-oriented interpretation:

- (127) He may be here.
- a. ✓He may be here (right now).
 - b. ✓He may be here (tomorrow).

When we add 'then' this ambiguity disappears, with only the future oriented reading being accessible:³⁵

- (128) He may be here then.
- a. ✗He may be here (right now).
 - b. ✓He may be here (tomorrow).

In the sentence above, the modal can only refer to a future time. The same pattern holds for past tense modals, which without 'then' are three-way ambiguous between (i) a past tense reading, (ii) a counterfactual epistemic present tense one, and (iii) a future tense reading.

- (129) He could have been here.
- a. ✓He could have been here (yesterday)
 - b. ✓He could have been here (now).
 - c. ✓He could have been here (tomorrow).

Again, the present tense interpretation is excluded the moment we add 'then':

- (130) He could have been here then.
- a. ✓He could have been here (yesterday)
 - b. ✗He could have been here (now).
 - c. ✓He could have been here (tomorrow).

Thus, it seems that modals that are ambiguous between a present tense interpretation and some other interpretation (past or future) lose the former once 'then' is added. Thus, the 'then'-present puzzle seems to hold with modals too. The same holds for a present tense inside an if-clause, which is ambiguous between a present- and a future-oriented reading:

- (131) If he is here, . . .
- a. ✓If he is here now, . . .
 - b. ✓If he is here tomorrow, . . .

Again, once 'then' is added we lose the present tense reading:

- (132) If he is here then, . . .

³⁵We thank Cleo Condoravdi for pointing this out to us during WCCFL.

- a. ✗If he is here now, . . .
- b. ✓If he is here tomorrow, . . .

As a reviewer points out, given that these examples involve a matrix and not an embedded present, we would predict that non-SoT languages such as Hebrew should behave similarly to English. This is indeed the case according to our consultant (Yael Sharvit, p.c.). Consider the following example recreating (127):

- (133) hu alul lihyot kan
he may inf-be here
- a. ✓He may be here (right now).
 - b. ✓He may be here (tomorrow).

The modal is ambiguous between a present- and a future-oriented interpretation, just like English (127). And exactly like English, when *az* ‘then’ is added only the future-oriented interpretation is possible:

- (134) hu alul lihyot kan az
he may inf-be here then
- a. ✗He may be here (right now).
 - b. ✓He may be here (tomorrow).

Moving on to past tense modals, again like in English they are three-way ambiguous between a past, present, and future interpretation.

- (135) hu alul haya lihyot kan
hu may past-be inf-be here
- a. ✓He could have been here (yesterday)
 - b. ✓He could have been here (now).
 - c. ✓He could have been here (tomorrow).

Again, when *az* ‘then’ is added the present-oriented interpretation is blocked:

- (136) hu alul haya lihyot kan az
hu may past-be inf-be here then
- a. ✓He could have been here (yesterday)
 - b. ✗He could have been here (now).
 - c. ✓He could have been here (tomorrow).

Finally, the present inside an if-clause is ambiguous between a present- and a future-oriented reading, as in English.

- (137) im hu kan, . . .
if he here . . .

- a. ✓If he is here now, . . .
- b. ✓If he is here tomorrow, . . .

However, when *az* ‘then’ is added the present-oriented reading disappears.

- (138) im hu kan az, . . .
 if he here then . . .
- a. ✗If he is here now, . . .
 - b. ✓If he is here tomorrow, . . .

We thus see that the ‘*then*’-present puzzle also extends to a present under a modal, as we would predict, since our theory predicts incompatibility between any (semantically contentful) present tense and ‘*then*’.

Non-past vs. present

In our theory, we account for the ‘*then*’-present puzzle thanks to the perspectival presuppositions of the present and ‘*then*’, which when evaluated with the same perspective yield a contradiction. Crucially, we do not predict *any* morphological present to be incompatible with ‘*then*’, only those that semantically presuppose an overlap with the perspective.³⁶ For example, we already saw that the English present embedded under *will* is a deleted present, and is thus compatible with *then*. A similar phenomenon also exists in Modern Greek in the form of a semantically nonpast, morphologically present tense. In our system, the non-past is correctly predicted to be compatible with ‘*then*’. Subjunctive mood in Modern Greek is marked with the *na* particle, such as in the following sentence:

- (139) I Maria theli na jrapsi ena vivlio.
 DET Maria want.PRES SUBJ write.PERF.NONPAST a book
 ‘Maria wants to write a book.’

Giannakidou (2009) noted that the tense on the embedded verb is a perfective nonpast, even though morphologically it surfaces as a present tense. She characterizes this as “a dependent verbal form with no formal mood features: the perfective nonpast (PNP)” (see also Holton et al. (2012)). PNP is a dependent form, since it cannot appear in root clauses (except under a future modal), and it can only surface inside a subjunctive clause. There is also no imperfective equivalent of the nonpast.

As Giannakidou (2009) notes, the PNP should semantically be distinguished from the present (even though morphologically PNP surfaces as a present). For our purposes, this means that PNP has present tense morphology without present tense semantics, i.e., without presupposing an overlap with the temporal perspective. Thus, we would predict it to be compatible with ‘*then*’. Indeed, this prediction is borne out:³⁷

³⁶We also do not *require* present tense morphology for the puzzle to arise. If another morphological tense is a present in disguise having the same presupposition, we also predict incompatibility with ‘*then*’.

³⁷We thank Anastasia Giannakidou for pointing this out to us during CLS.

- (140) To 2000, o Yanis ithele i Maria na mini egkios tote.
DET 2000 DET Yanis want.PAST DET Maria SUBJ get.PNP pregnant then
'In 2000, Yanis wanted Maria to get pregnant then.'

We thus see that the 'then'-present puzzle is a puzzle about a semantically (and not simply a morphologically) present tense. Our theory correctly predicts this.